THE CURE CENTER
FOR ULTRASOUND RESEARCH & EDUCATION

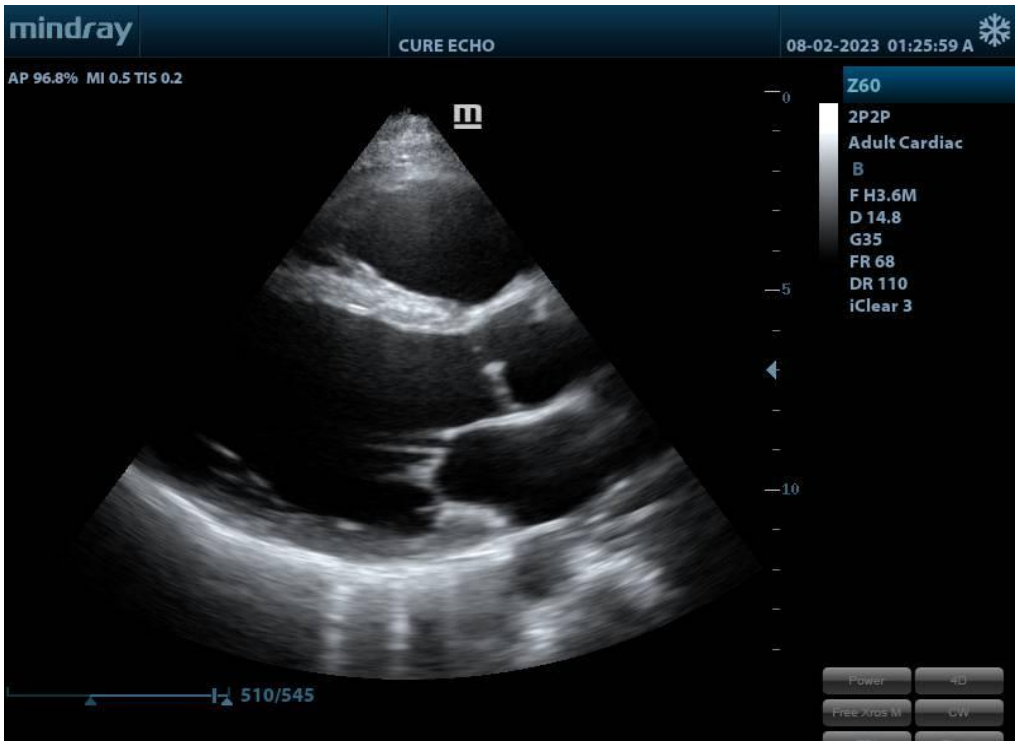
CURE Adult Echocardiography Protocol

***Clip: When an image is indicated as a clip that is a moving video clip that consist of 3 beats. To acquire a clip: Obtain clear image, count 3 heart beats, press freeze followed by save 2.**

Parasternal long axis (PLAX): *Patient is to lay in a left lateral decubitus position, with the left arm under patient' s head and right arm resting on the patient' s right hip. Notch should be indicated towards the patient's right shoulder (11oclock). When finding window, start at the 2nd intercostal space with the probe between the sternum and the breast tissue and move down the sternum until optimal image has been achieved.*

1. **2d Clip**

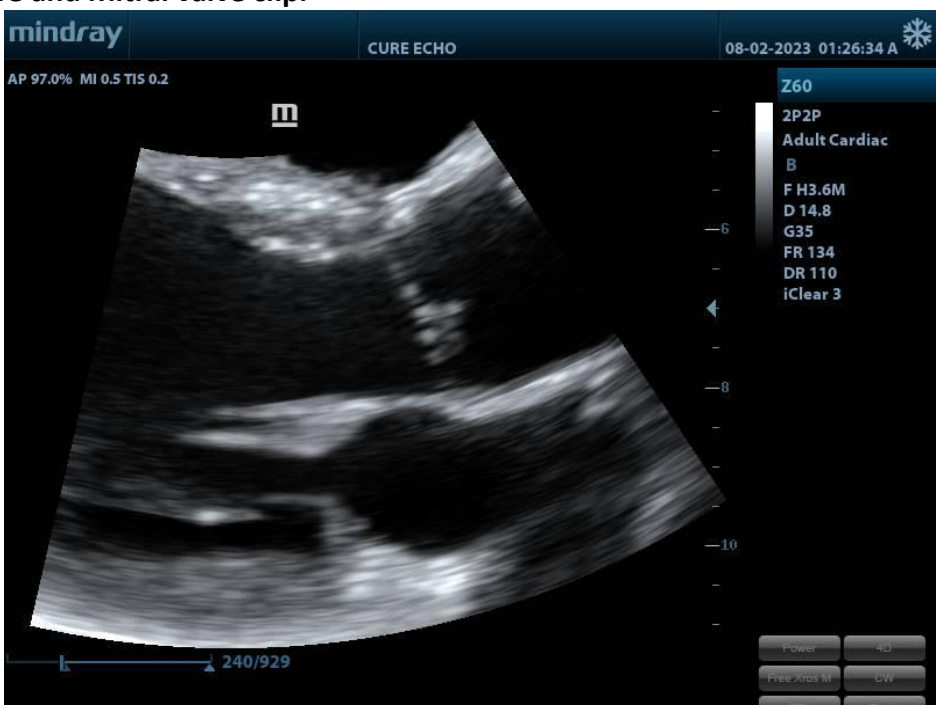
Note: Right Ventricle, Interventricular septum, left ventricle, left atrium, aortic valve, mitral valve should all be clearly visualized. Posterior to the heart the descending thoracic aorta should also be visualized. All chambers should be anechoic, and all tissues should be hyperechoic.



2. **Zoom Aortic valve and Mitral valve clip.**

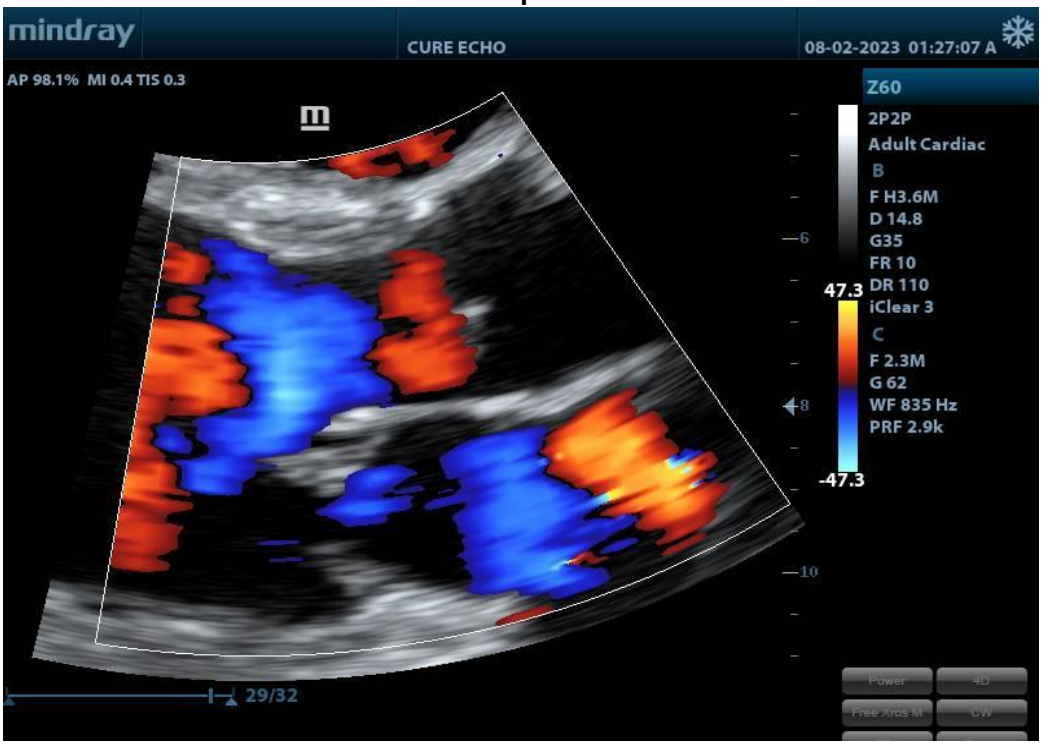
Note: Both valves and corresponding leaflets should be visible in this image to aid in identifying structural abnormalities.

*To zoom press the center of the depth wheel and adjust box size then press again



3. **Zoom and Color Aortic valve and Mitral valve clip.**

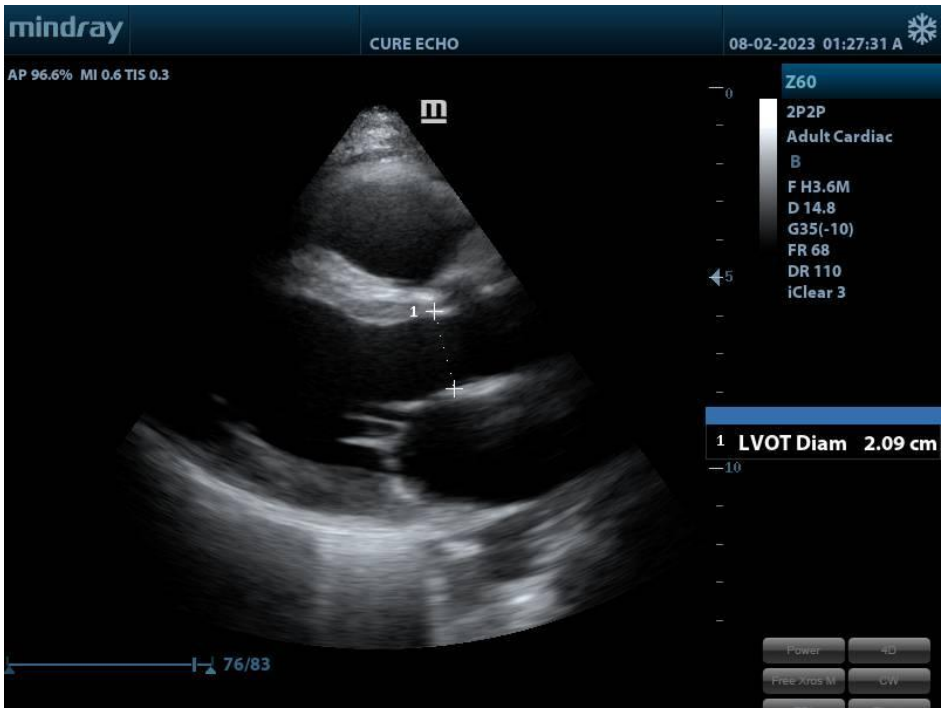
Note: Color box should be large enough that both mitral and aortic blood flow is visible without compromising frame rate.
*Exit zoom for the next images



4. Left ventricular outflow tract (LVOT) diameter

Note: Left ventricular outflow tract is located just before the aortic valve. During this the aortic valve should be open and should be measure inner to inner.

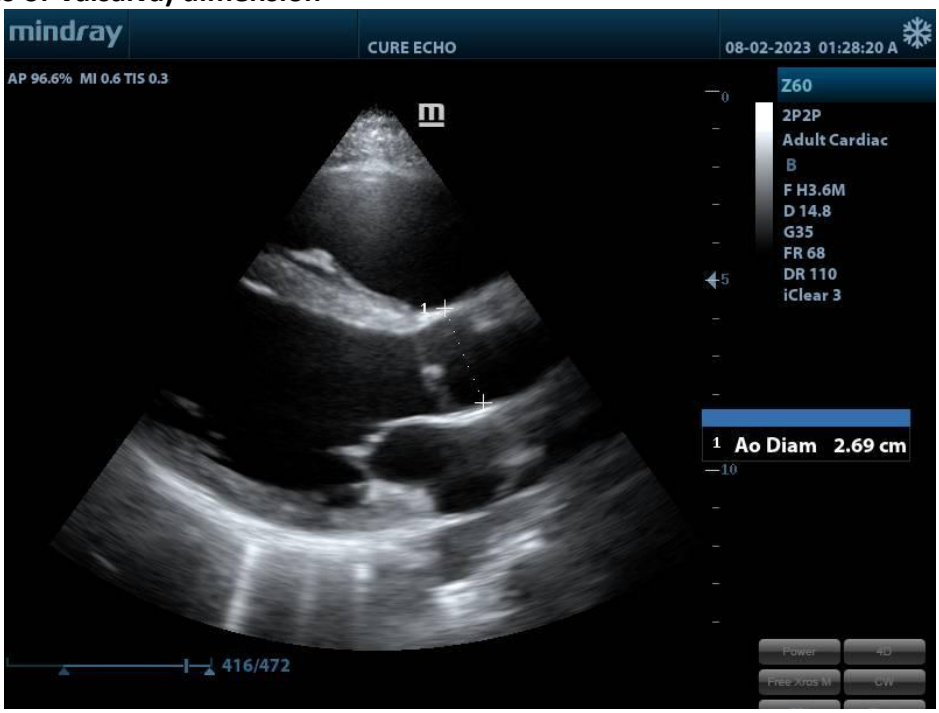
Click measurement, LVOT diameter and set points.



5. Aortic root (Sinus of Valsalva) dimension

Note: Aortic root diameter should be measured when the aortic valve is closed, using leading edge to leading edge measurements (outer to inner).

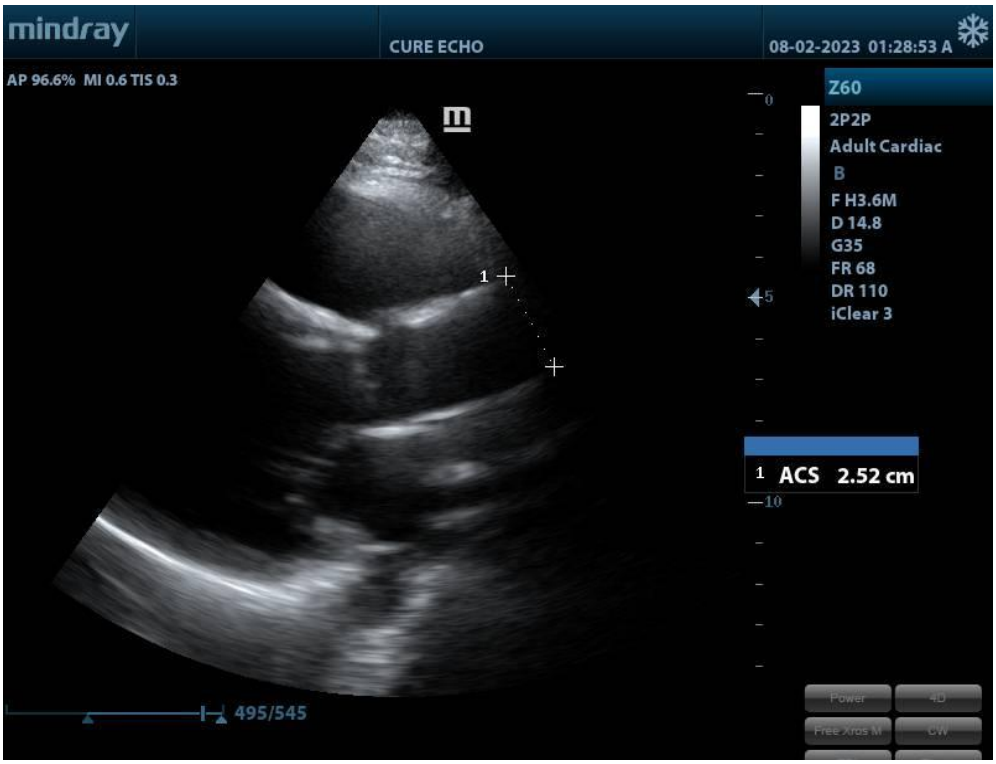
Click measure, AO diam and set points.



6. Ascending Aorta Diameter

Note: Diameter measurements should be taken from the most distal part of the ascending aorta, leading edge to leading edge (outer to inner).

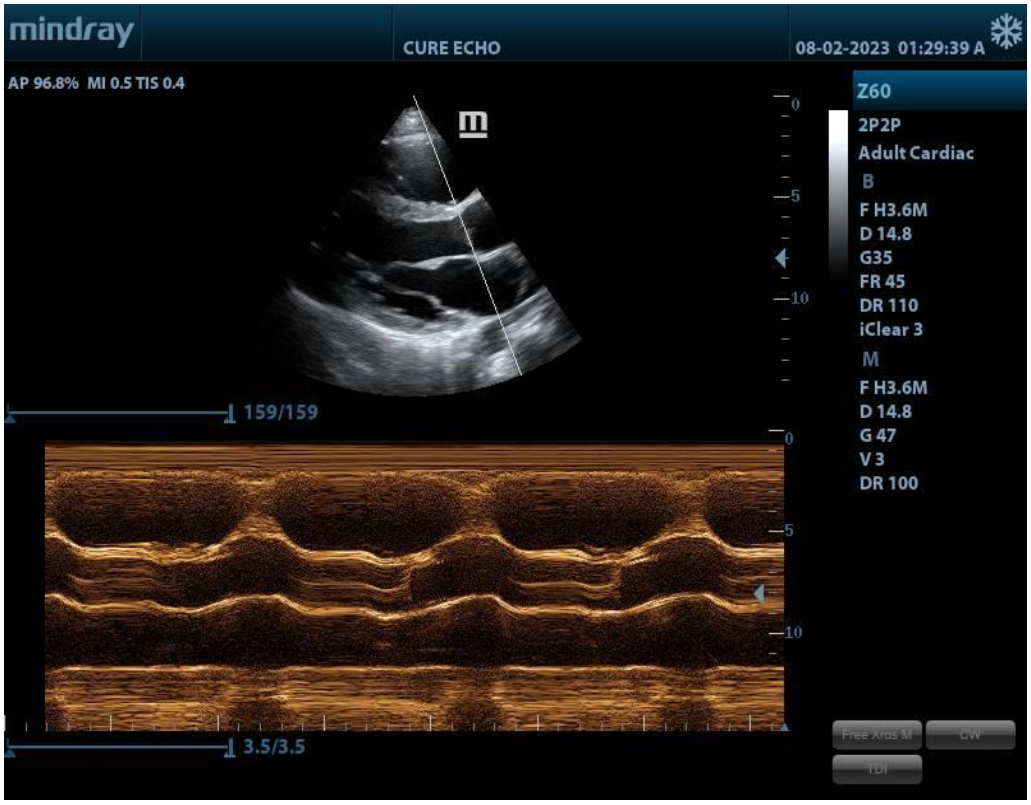
Click measure, ACS, and set points.



7. M-Mode @ Aortic valve level (AOV)

Note: M-mode cursor is aligned perpendicular to the aortic valve capturing both RCC and NCC motion. Valve opening and valve closure should be identifiable.

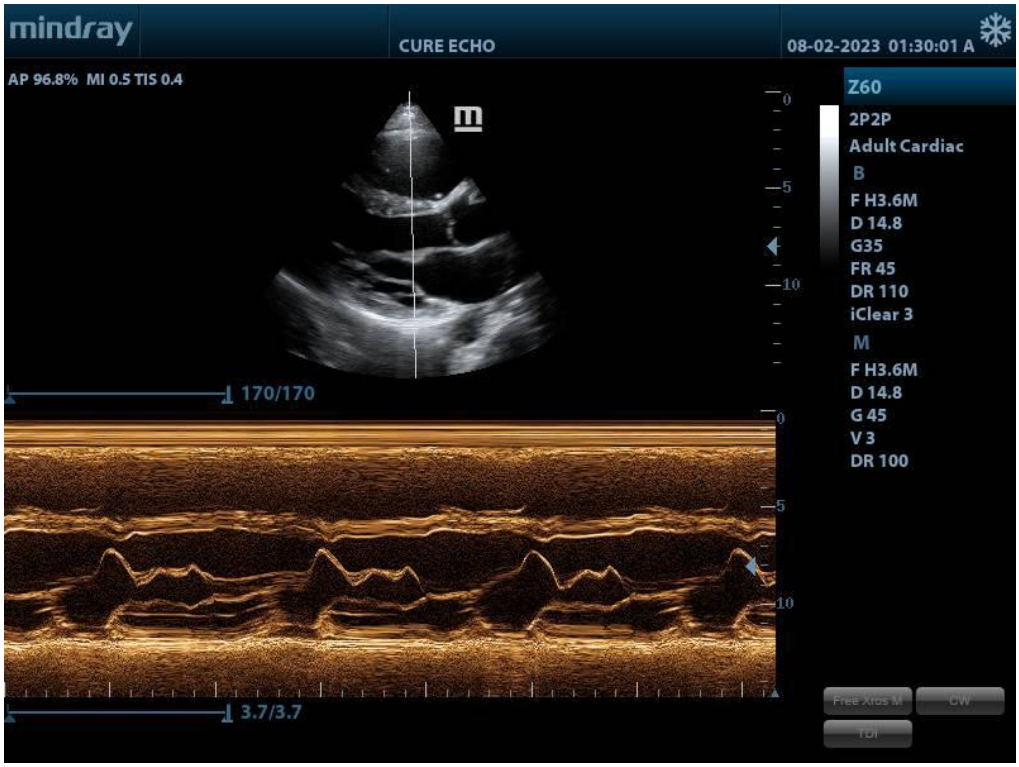
RCC= Right coronary cusp
NCC= Left coronary cusp



8. M-mode @ mitral valve level (MV)

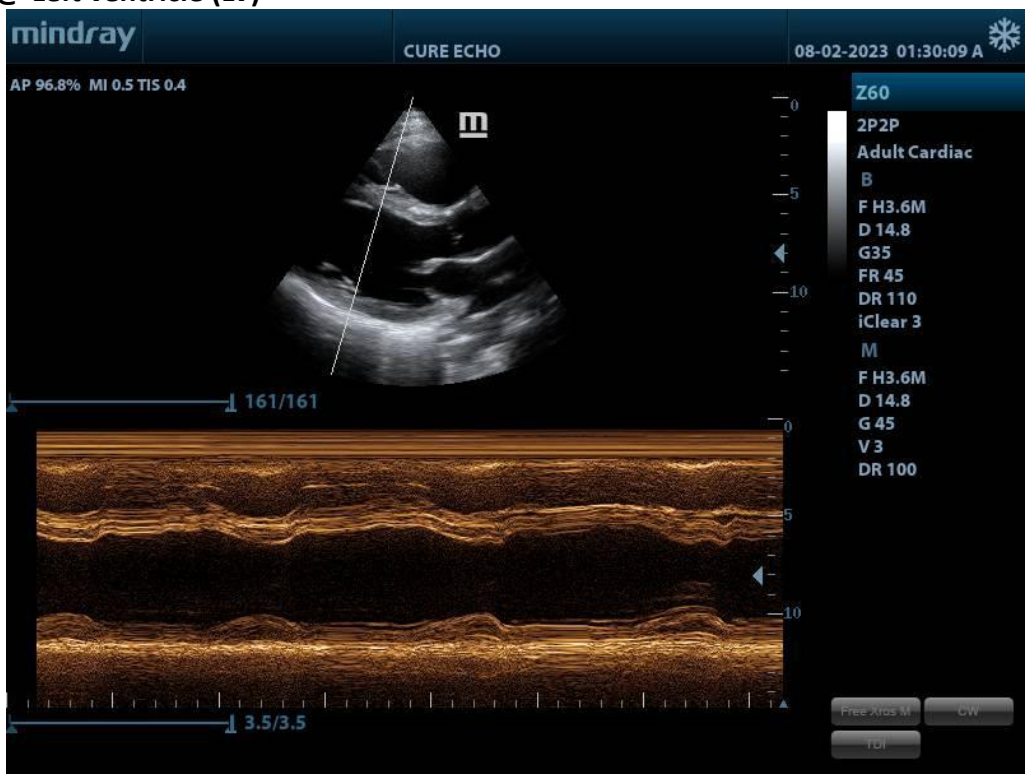
Note: M-mode cursor is aligned perpendicular to the mitral valve capturing both AMVL and PMVL motion. Valve opening and valve closure should be identifiable.

AMVL= Anterior mitral valve leaflet
PMVL= Posterior mitral valve leaflet



9. M-mode @ Left ventricle (LV)

Note: M-mode cursor is aligned perpendicular to the left ventricle capturing both IVS and LVPW (endocardium)motion. Systolic thickening of septum and endocardium should be identifiable.



10. Left ventricular interior dimension in diastole measurement (LVIDd).

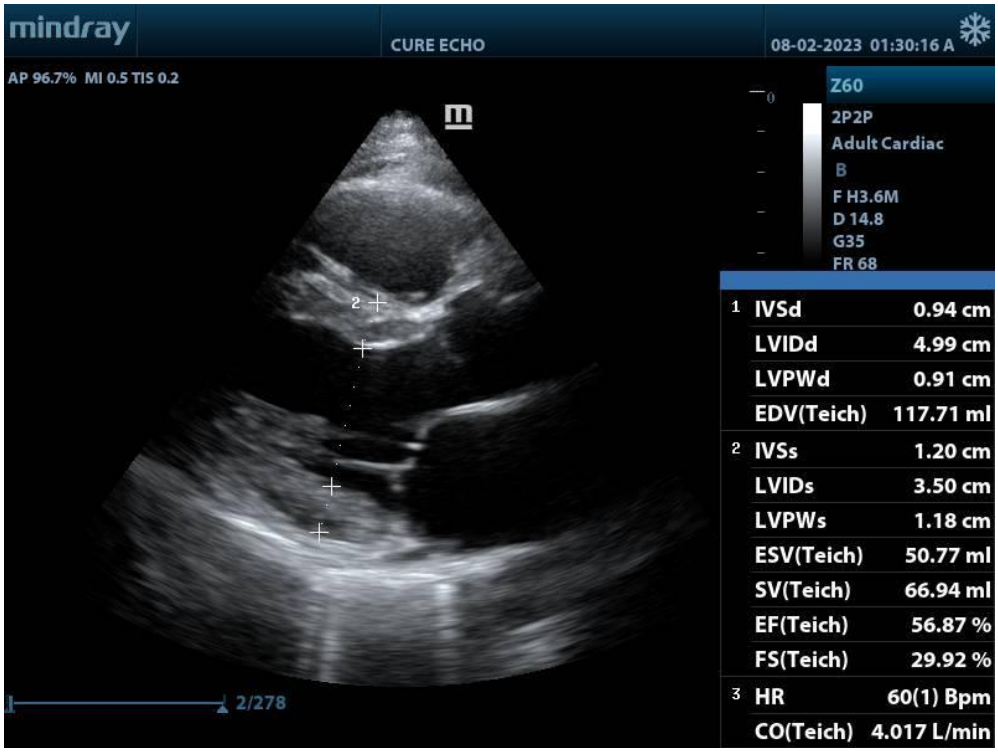
Note: Teicholz measurement- With the chamber in **diastole** with the mitral and aortic valves closed, Chamber measurements should be taken. Click measure, Teicholz, diastole and begin setting points starting with:IVS, LV, LVPW.



11. LVIDs Measurement

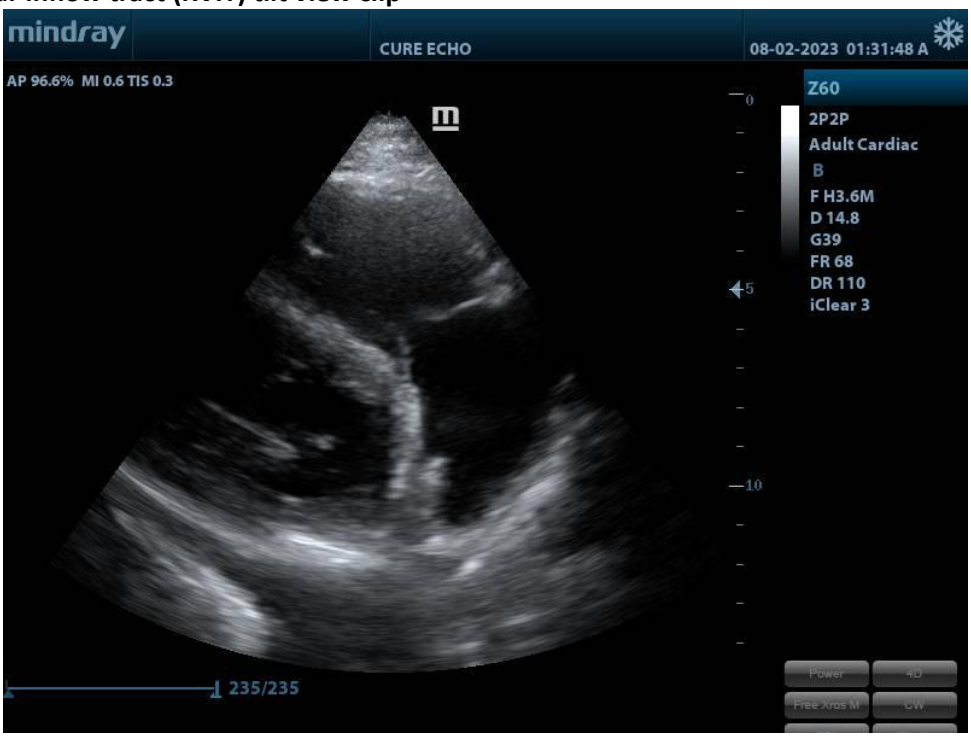
Note: Teicholz measurement- With the chamber in **systole** with the mitral and aortic valves closed, Chamber measurements should be taken. Click measure, Teicholz, diastole and begin setting points starting with: IVS, LV, LVPW. When prompted to add Heart rate enter 60 as a default unless you are able to determine patients HR

PAY ATTENTION TO EF



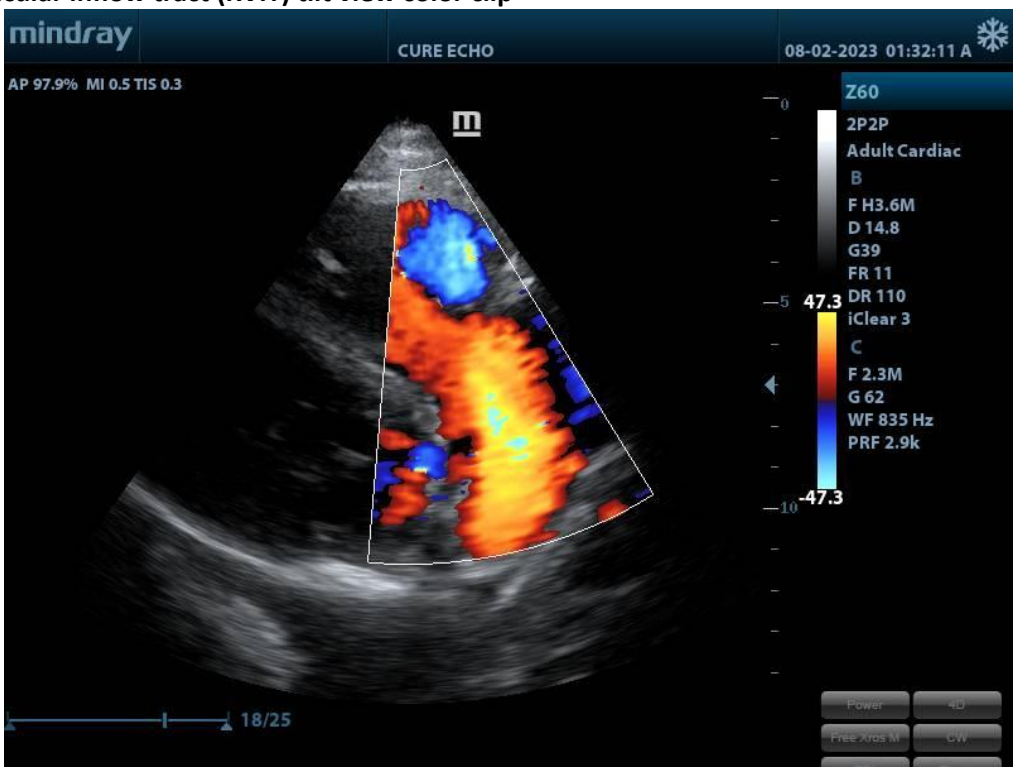
12. Right ventricular inflow tract (RVIT) tilt view clip

Note: With the same PLAX orientation (11 o'clock) angel inferiorly toward the patients hips to open the right ventricular inflow tract. The RVIT consist of : Right ventricle, tricuspid valve, right atrium



13. Right ventricular inflow tract (RVIT) tilt view color clip

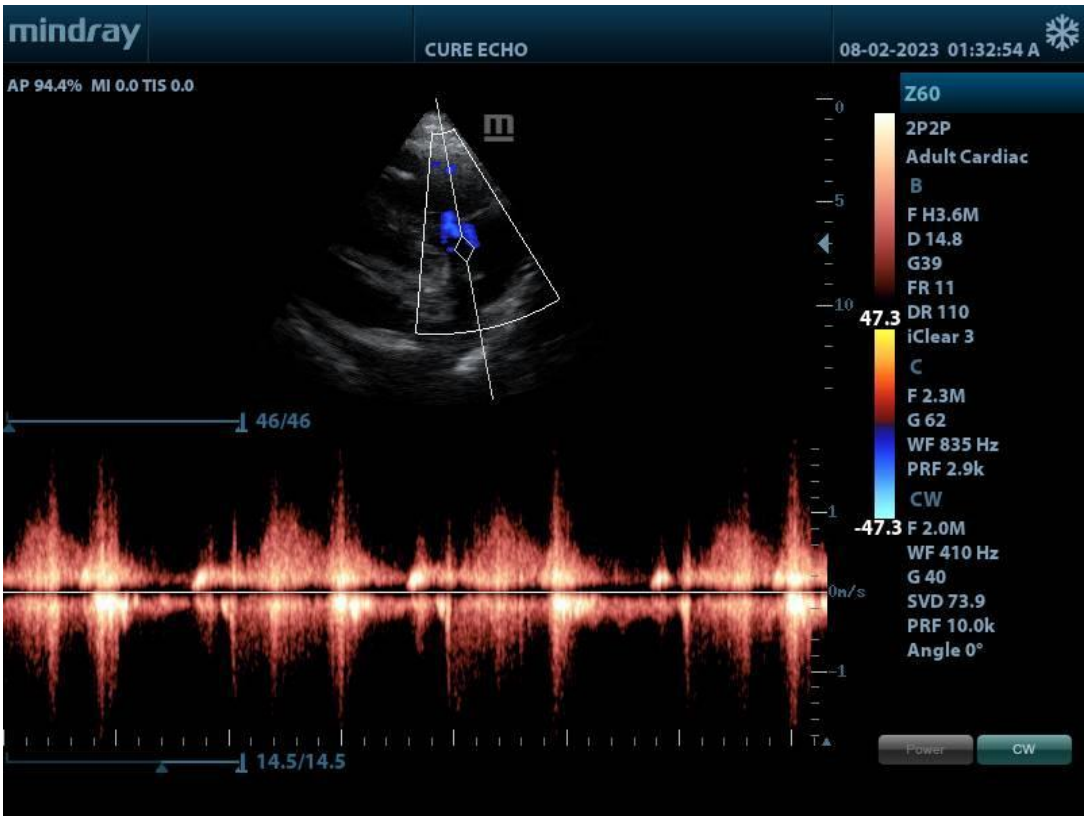
Note: Color box should be over the right ventricle, tricuspid valve, and right atrium to assess right ventricular inflow and ruling out tricuspid regurgitation. *Note the red inflow due to flow towards the transducer.



14. Continuous Wave Doppler RVIT

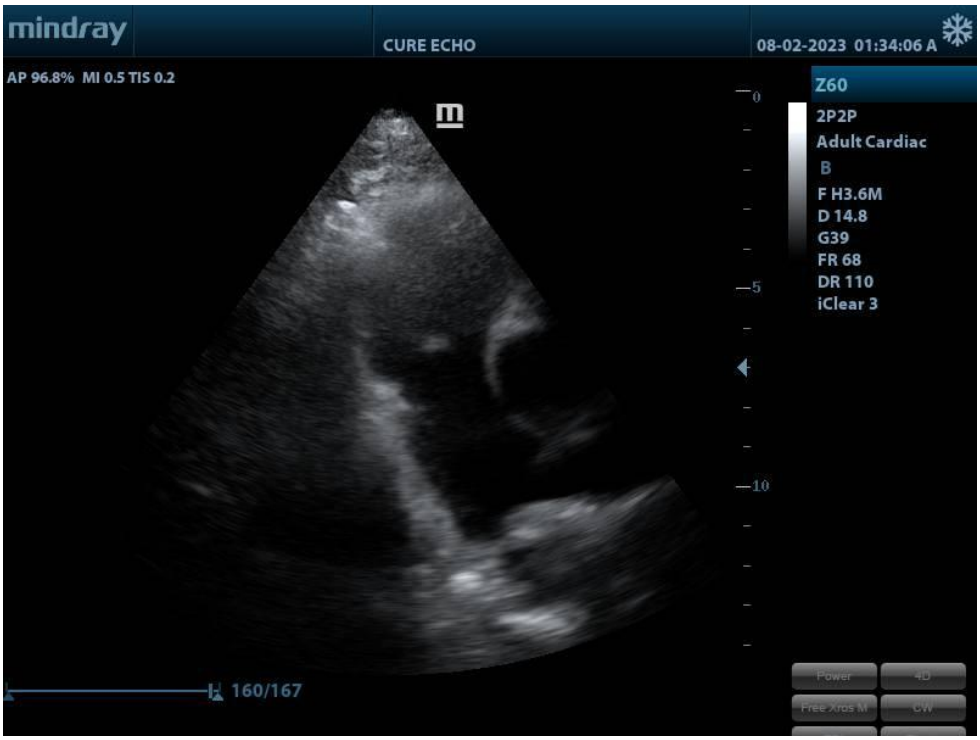
Note : Color box should be over the right ventricle, tricuspid valve, and right atrium and CW cursor should be positioned parallel to the blood flow within the chamber. Note the normal RVIT flow is a positive doppler shift (above the baseline)

*hit the print button to deploy CW cursor, algin with flow and hit print again to begin doppler



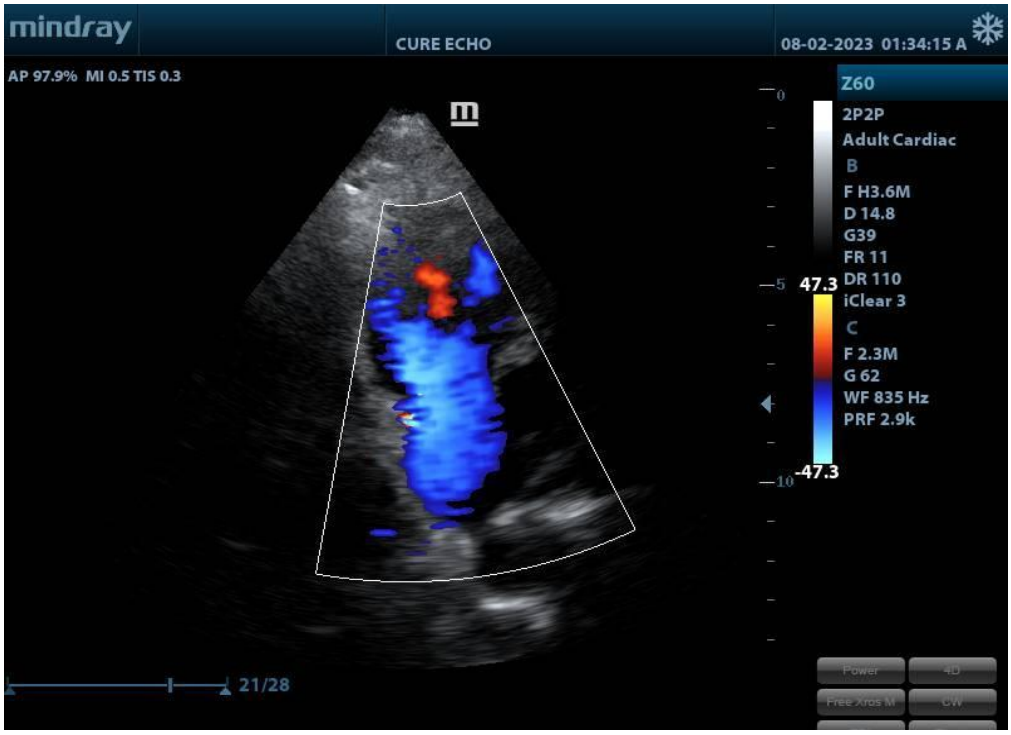
15. Right ventricular outflow tract (RVOT) clip

Note : With the same PLAX orientation (11 o'clock) angel superior towards the head to open the right ventricular outflow tract. The right ventricular outflow tract consists of the Right ventricle, pulmonic valve and pulmonary artery



16. Right ventricular outflow tract (RVOT) color clip

Note: Color box should be over the right ventricle, pulmonic valve, and main pulmonary artery to assess right ventricular outflow and ruling out pulmonic regurgitation. *Note the blue outflow due to flow away the transducer.

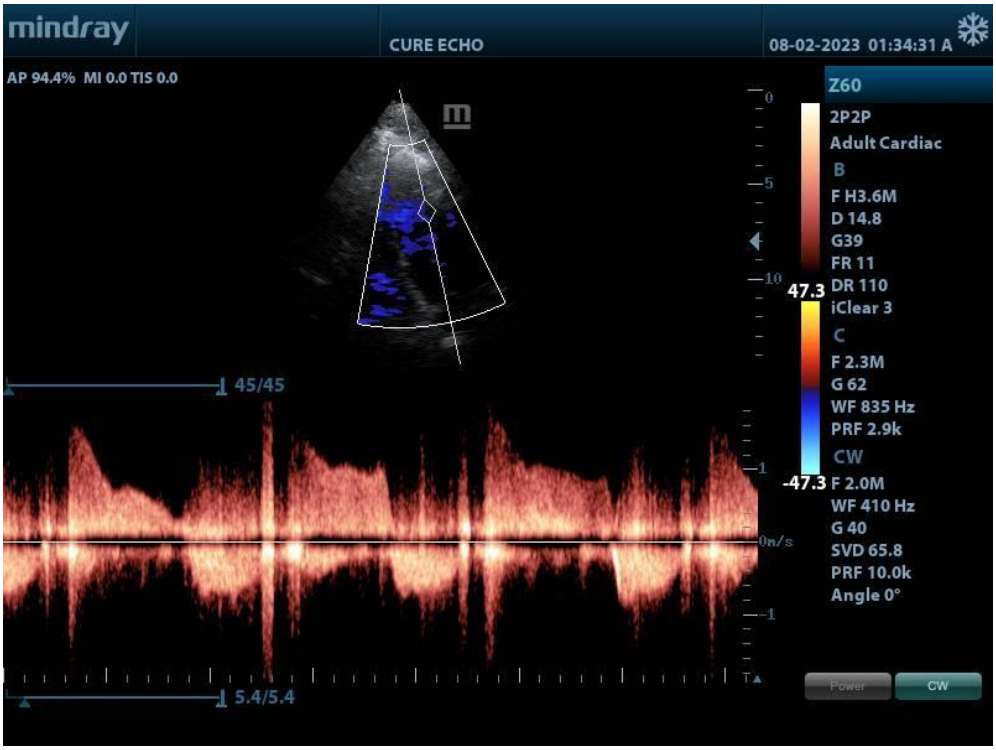


17. Continuous Wave Doppler RVOT

Note: CW cursor should be positioned parallel to the blood flow within the chamber. Note the normal RVOT flow is a negative doppler shift (below the baseline)

*Hit the print button to deploy CW cursor, align and hit print to begin doppler

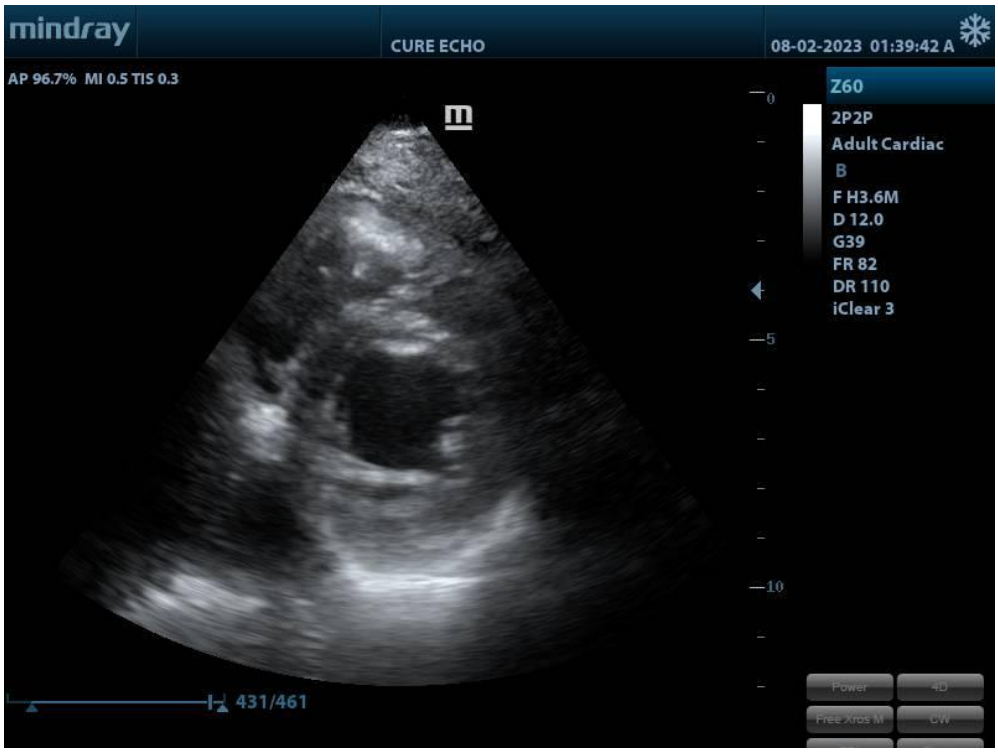
*This example also demonstrates pulmonic regurgitation seen above the baseline



Parasternal short axis (PSAX): *Notch should be rotated to the patients left shoulder (1 o'clock). This will be your cross-sectional views of the left ventricle.*

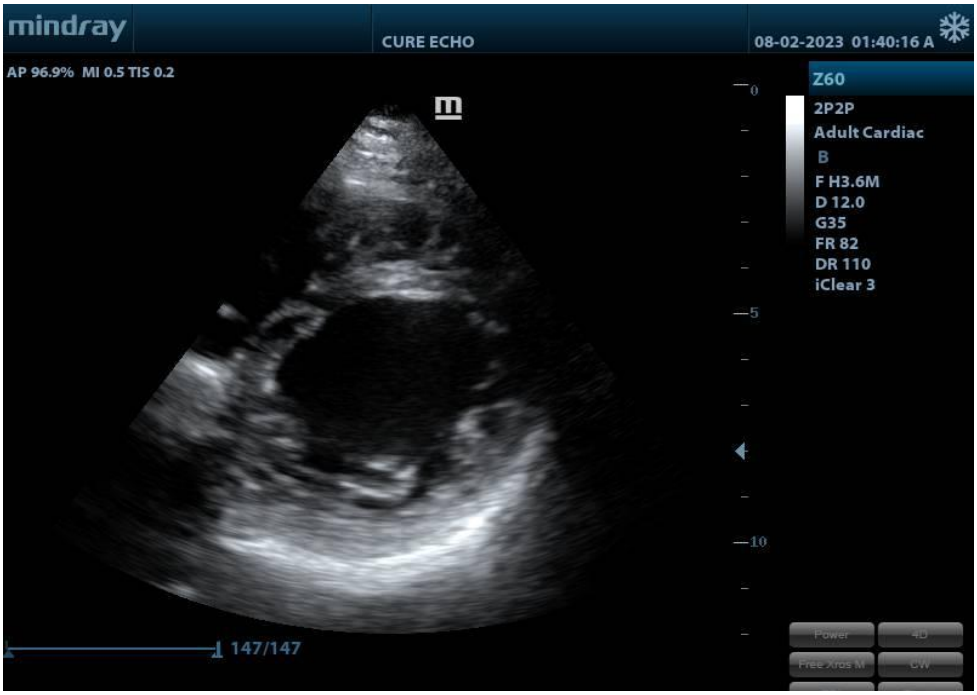
18. Apex clip

Note: With notch at 1oclock the probe should be angled inferiorly towards patient's hip. The LV apex will show a small LV cavity.



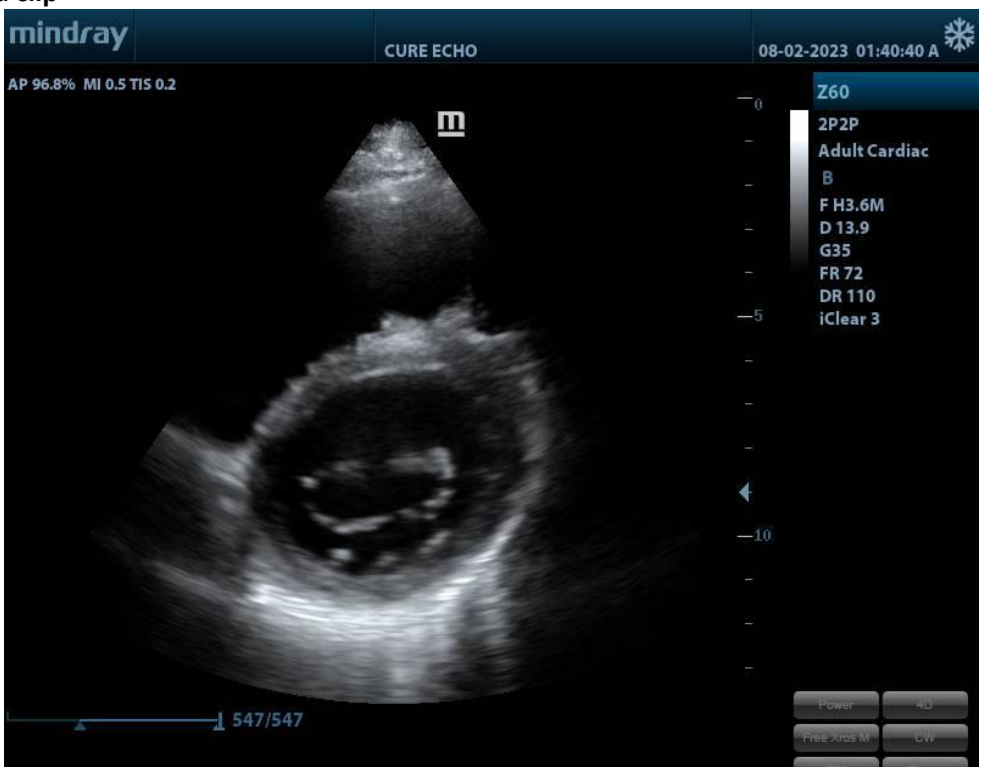
19. Papillary muscle clip

Note: With notch at 1oclock the probe should be slightly manipulated upwards away from the hips. LV size should increase, and papillary muscles should be visual. *Make very small movements*



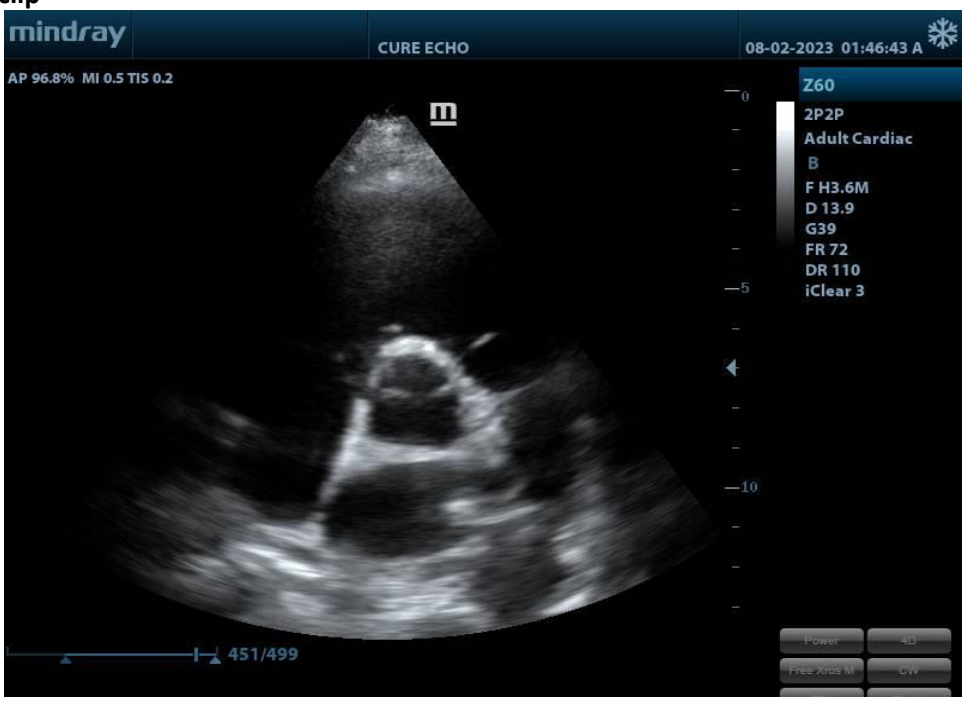
20. Mitral valve 2d clip

Note: With notch at 1oclock the probe should be slightly manipulated upwards away from the hips. AMVL and PMVL should be visible.
Make very small movements



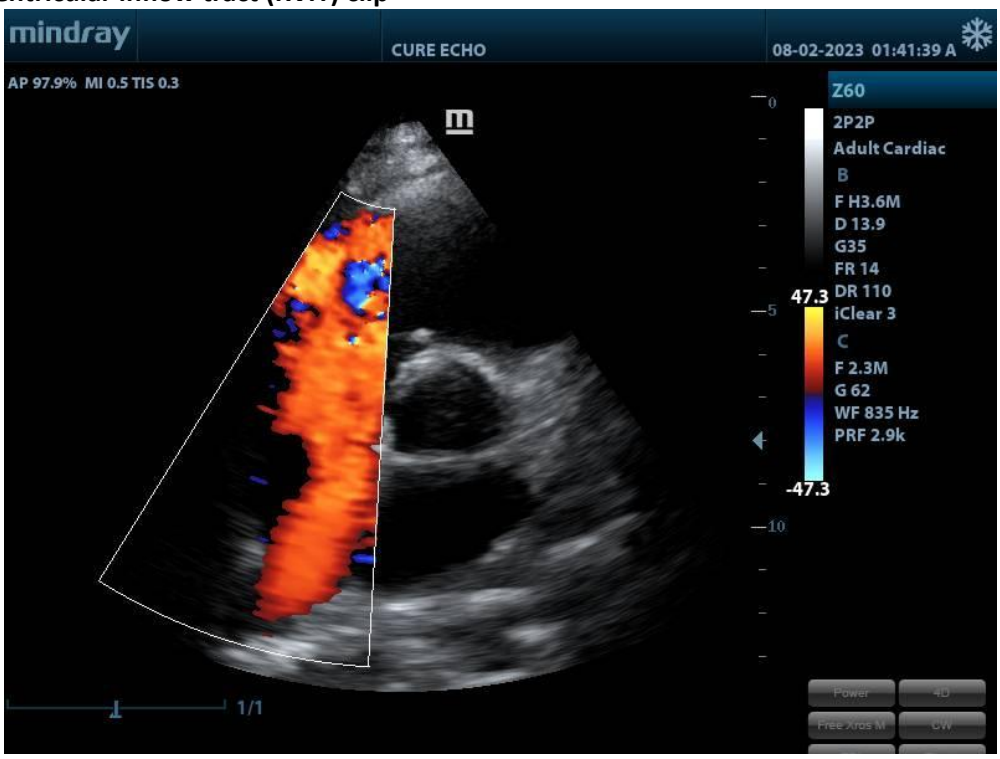
21. Aortic valve 2d clip

Note: With notch at 1oclock the probe should be angled up towards the patients head. Tricuspid valve, aortic valve, pulmonic valve, and left atrium should all be visible. Aortic valve should show all 3 leaflets in this view “Mercedes symbol”.



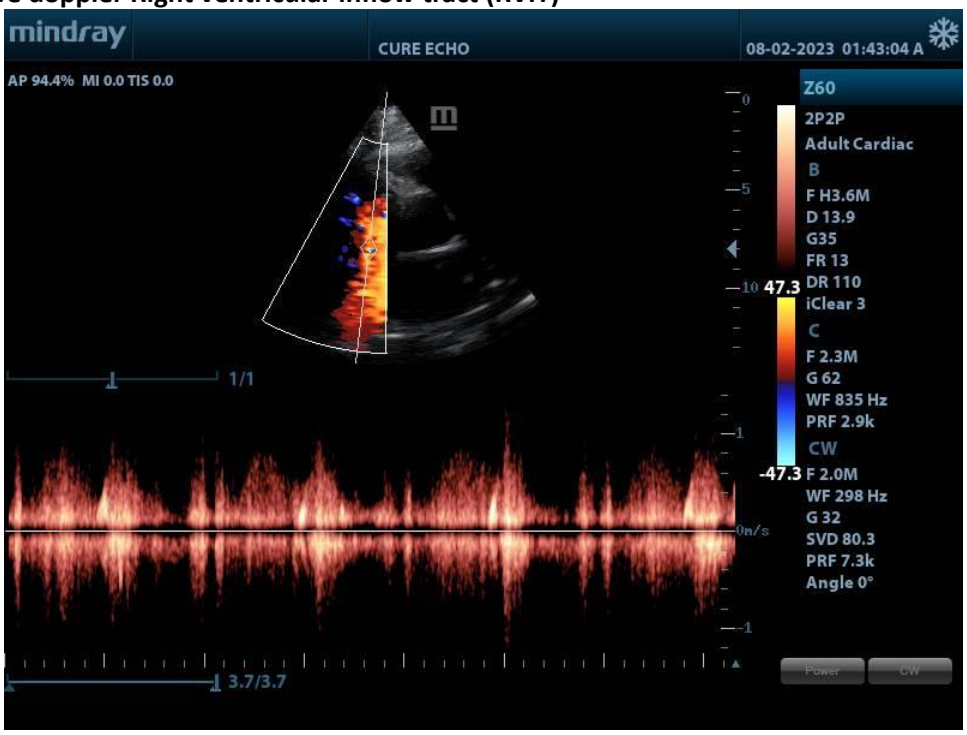
22. Color right ventricular inflow tract (RVIT) clip

Note: With notch at 1oclock the probe should be angled up towards the patient head. Tricuspid valve, aortic valve, pulmonic valve, and left atrium should all be visible. Color box should be over the RVIT to rule out tricuspid regurgitation.



23. Continuous wave doppler Right ventricular inflow tract (RVIT)

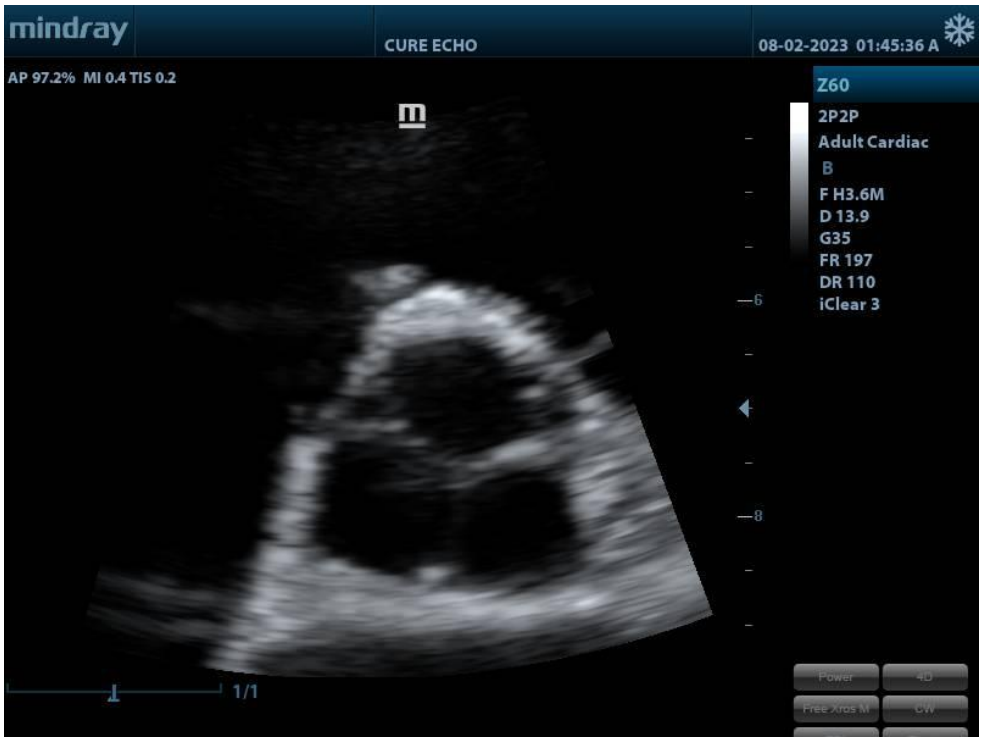
Note: CW doppler should be placed in the middle of the valve for optimal assessment of RVIT flow. *In cases of tricuspid regurgitation, it is important to place CW cursor parallel to TR jet*
*Hit the print button to deploy CW cursor, align and hit print to begin doppler



24. Zoom aortic valve clip.

Note: Zoom box should be large enough to capture entirety of aortic valve. All three leaflets should be visualized. *Since leaflets are so thin, we use zoom to optimize axial resolution*

*To zoom press the center of the depth wheel and adjust box size then press again

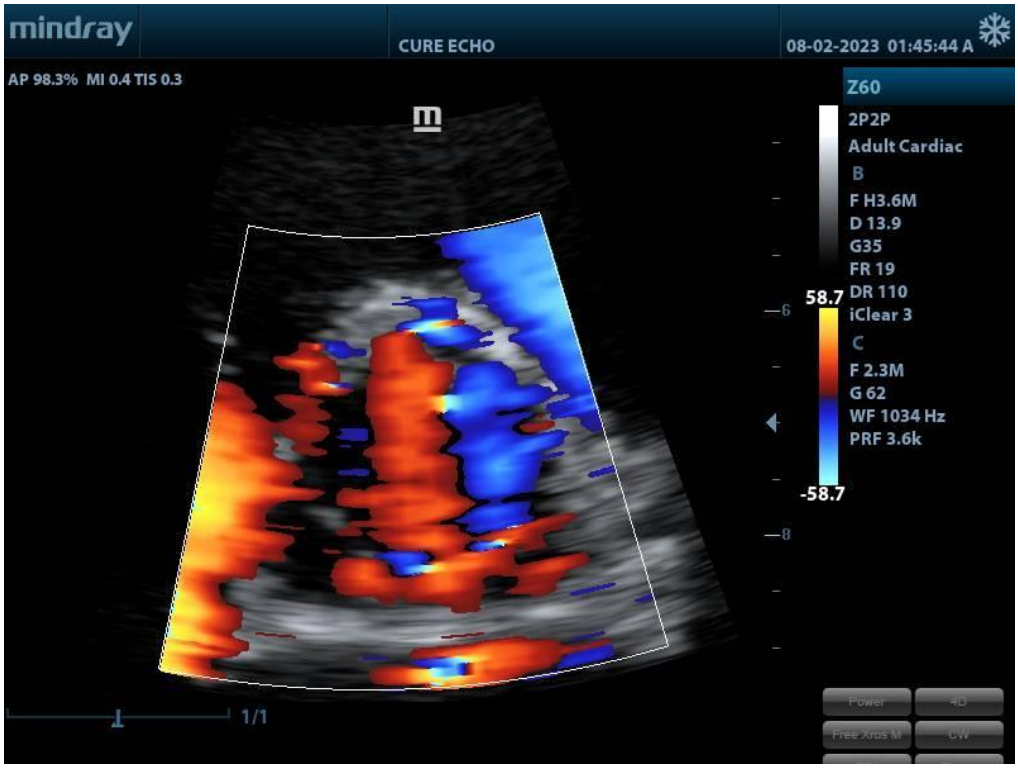


25. Zoom aortic valve with color clip.

Note: With zoom box in the same position, color should be added to rule of the presence of aortic regurgitation.

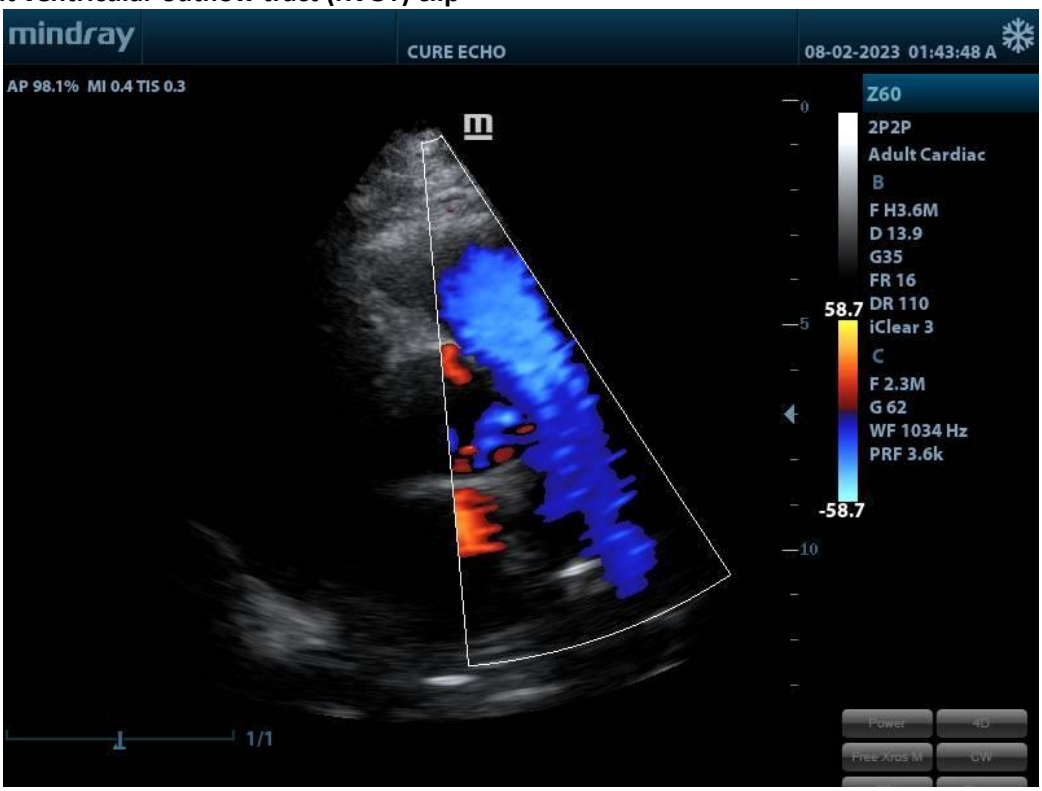
*To zoom press the center of the depth wheel and adjust box size then press again

*exit zoom for the next images



26. Color right ventricular outflow tract (RVOT) clip

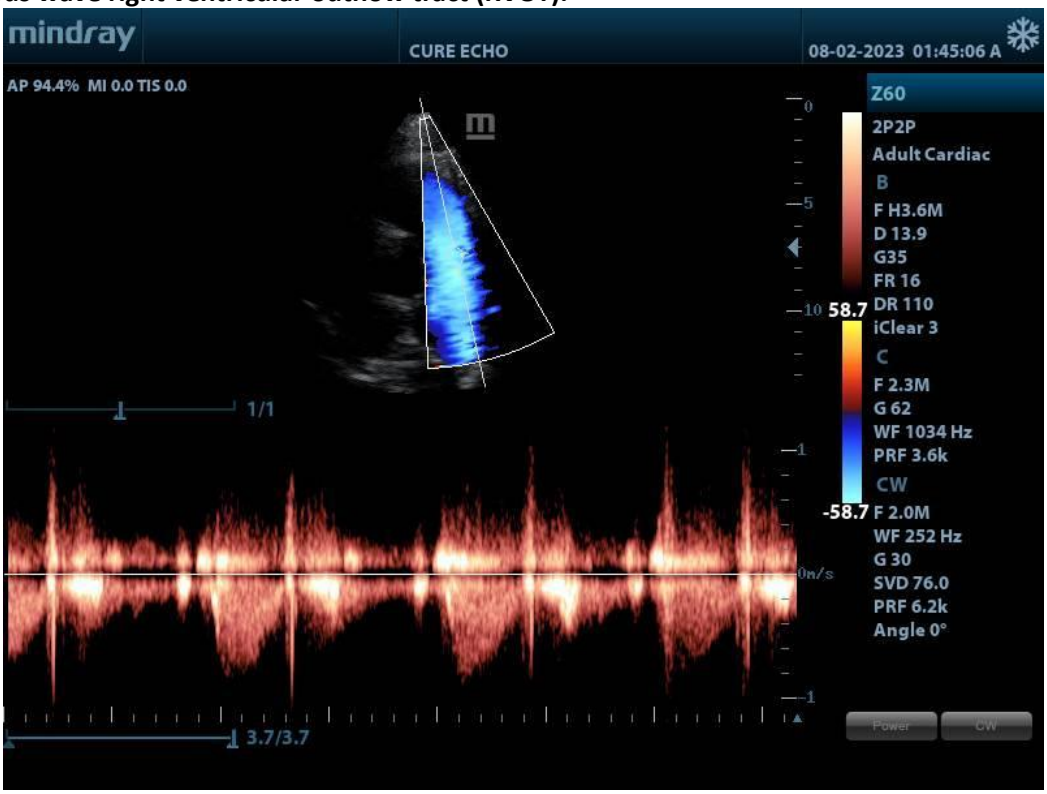
Note: With notch at 1oclock the probe should be angled up towards the patient's head. Tricuspid valve, aortic valve, pulmonic valve, and left atrium should all be visible. Color box should be over the RVOT to rule out pulmonic regurgitation.



27. Continuous wave right ventricular outflow tract (RVOT).

Note: CW doppler should be placed in the middle of the valve for optimal assessment of RVOT flow. *In cases of pulmonic regurgitation, it is important to place CW cursor parallel to PR jet*

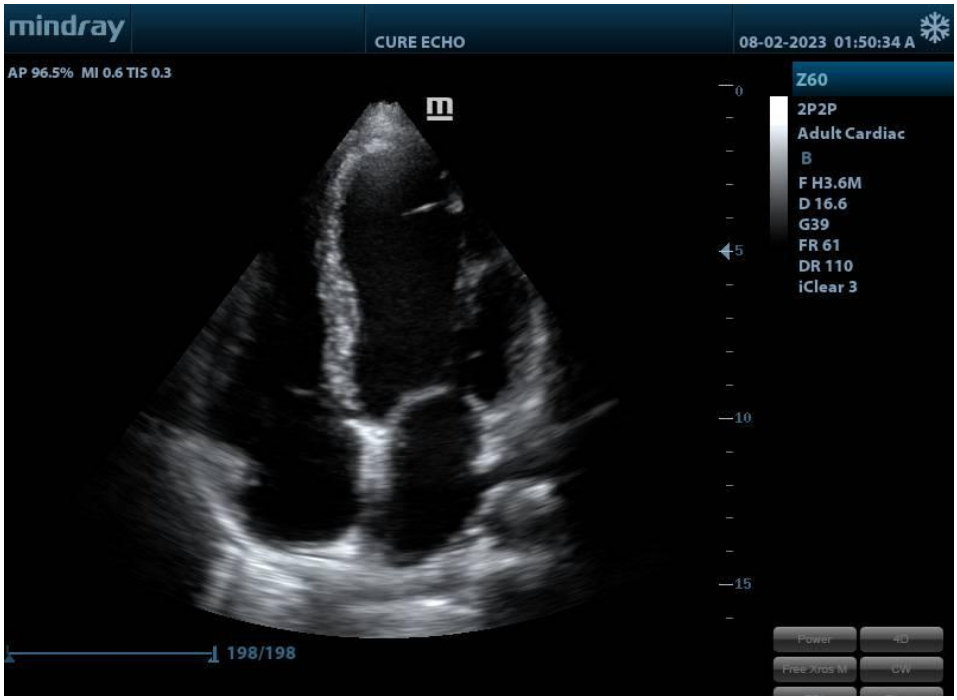
*Hit print button to deploy CW cursor, align, print again to begin doppler



***APICAL VIEW**: Probe should relocate to the apical position in the 5th intercostal space. Starting with a lateral approach and sweeping medially until all 4 chambers are clearly visible without foreshortening the apex of the heart (cutting-off the tip of the apex).*

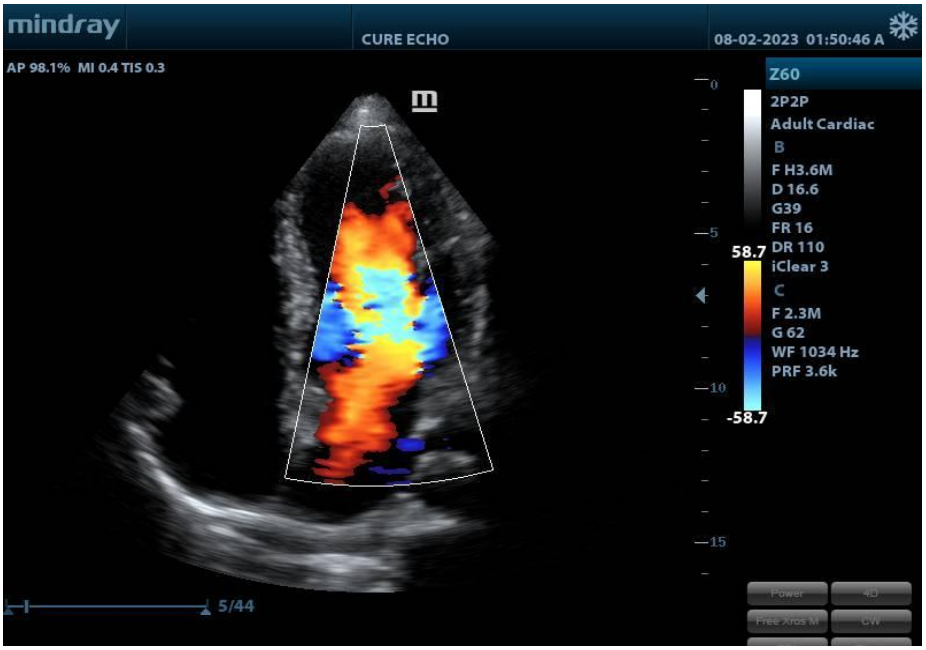
28. Apical 4 chamber clip

Note: All 4 chambers are clearly visible. Apex of the heart is visible (apex is smaller than rest of cavity). IVS should be parallel to the transducer marker (ventricle straight up and down).



29. Color LVIT clip

Note : All 4 chambers are clearly visible. Color box is focused on the inflow tract of the heart. Rule out mitral regurgitation



30. Pulsed wave left ventricular inflow tract (LVIT).

Note: Using PW the doppler cursor should be parallel to the flow and placed in the middle of the mv with the sample volume at the tips of the mitral valve leaflets. Normal LVIT flow will appear as a positive doppler shift.

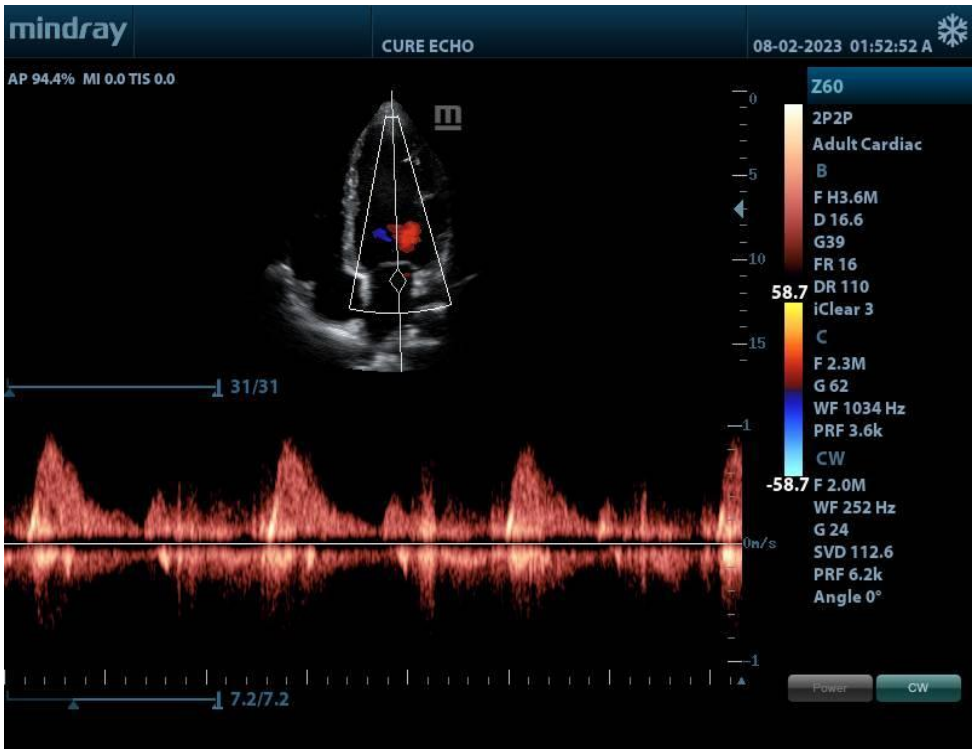
*Hit PW, align cursor, hit pw
*measure, mitral valve, MV peak E, MV peak A, MV DecT



31. Continuous wave left ventricular inflow tract

Note : Using CW cursor should be aligned parallel to the lvit flow. Normal lvit flow should remain a positive doppler shift while regurgitation will appear as a negative shift.

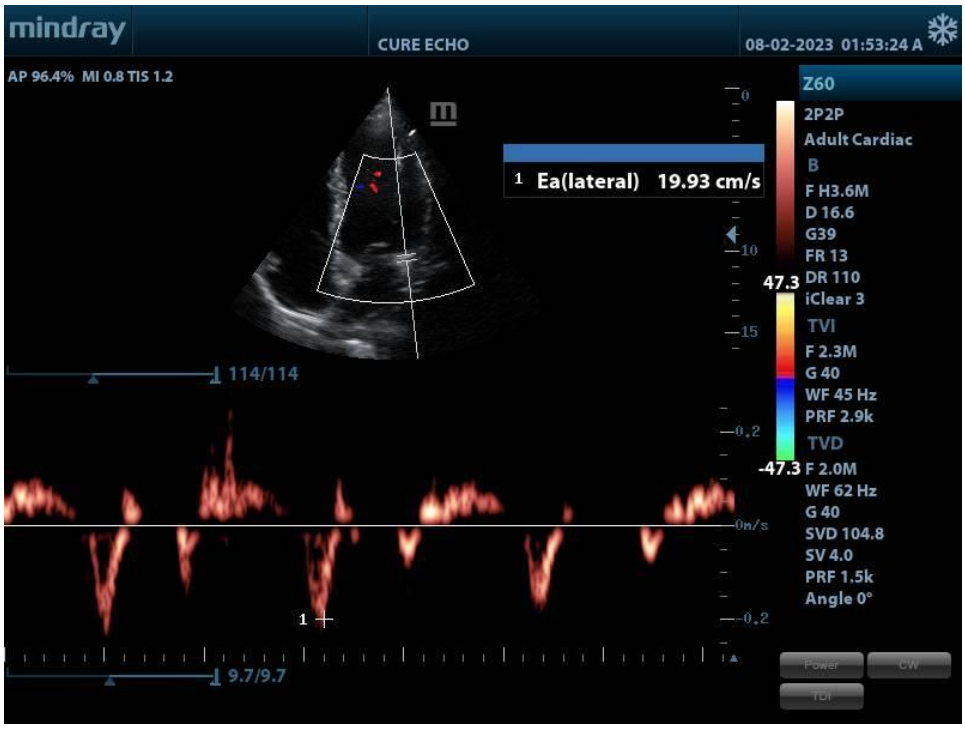
*Hit print to deploy doppler, align, print to begin doppler



32. Tissue doppler index (TDI) lateral annulus

Note: Using TDI package color box should be over both lateral and medial mv annulus. Place PW sample volume over the lateral mv annulus.

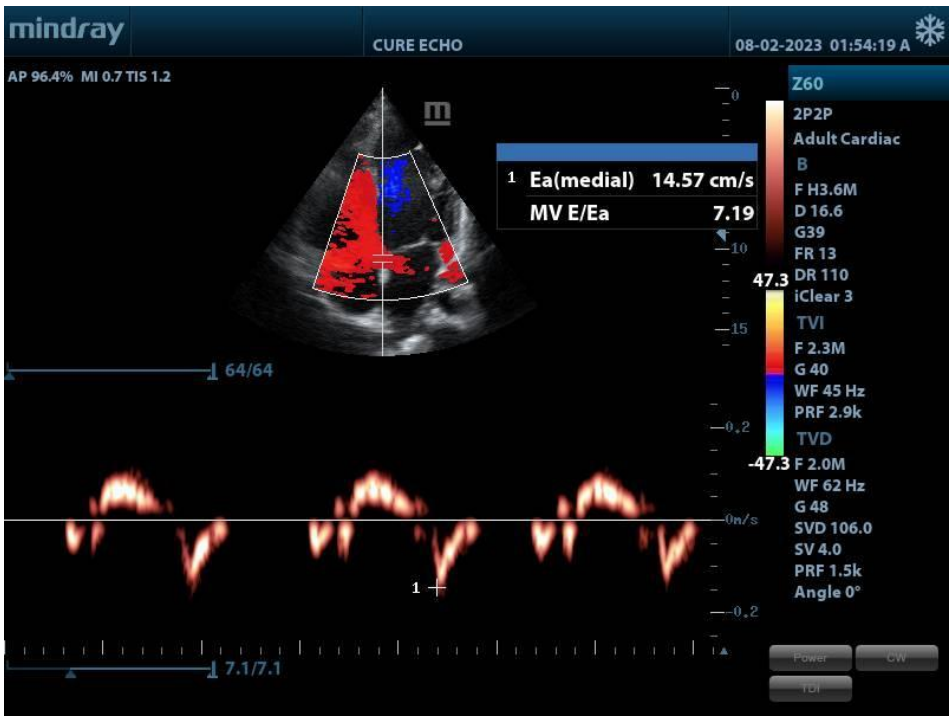
*Measure, tdi, Ea(lateral).



33. TDI medial annulus

Note: Using TDI package color box should be over both lateral and medial mv annulus. Place PW sample volume over the medial mv annulus.

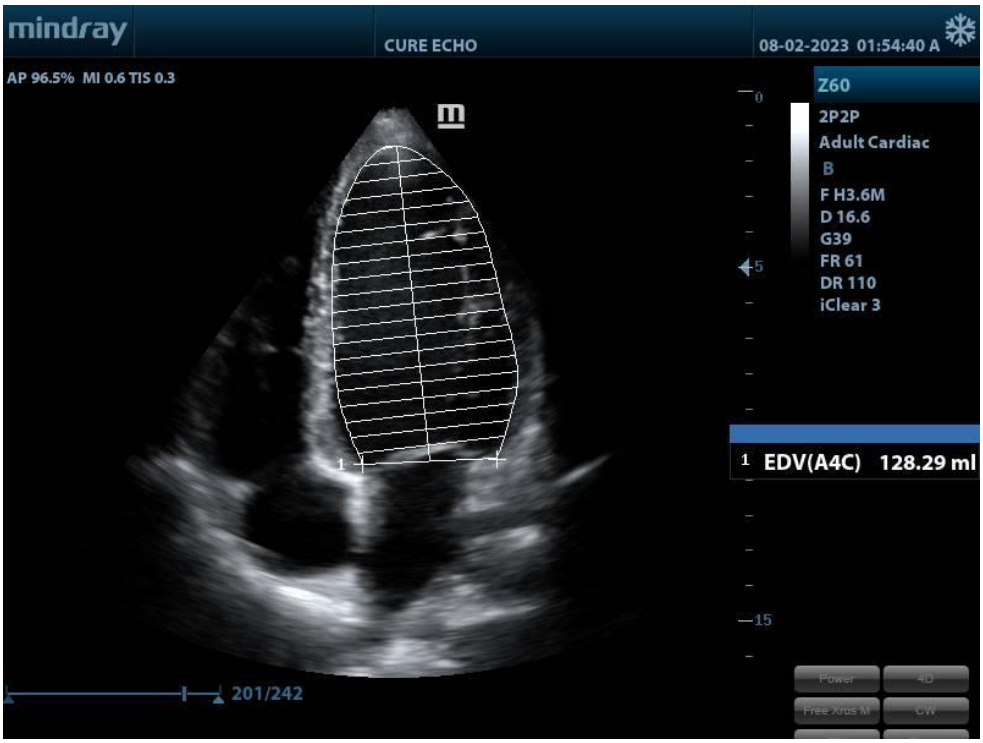
*Measure, tdi, Ea(medial)



34. BI-plane Simpson's A4C diastole

Note: While the LV is at an end diastolic state (largest dimension) measure EDV. Starting at the medial annulus, place points tracing the endocardium to the apex and back down to the lateral annulus.

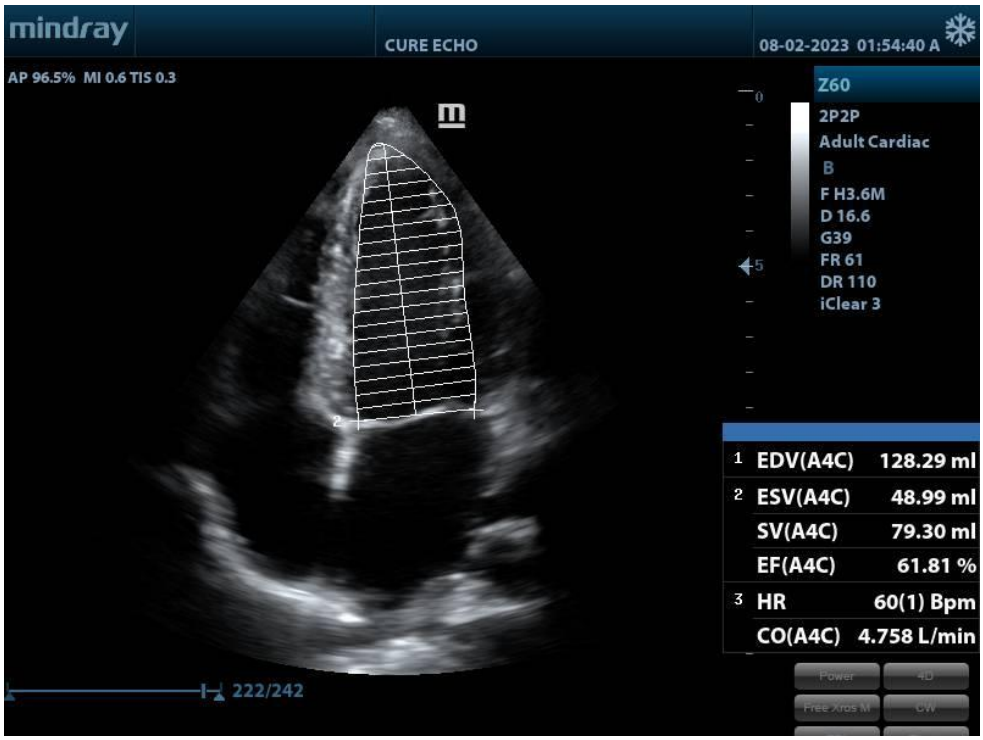
*Measure, Simpson, EDV(A4C).



35. Bi-plane Simpson's A4C systole

Note: While the LV is at an end systolic state (smallest dimension) measure ESV. Starting at the medial annulus, place points tracing the endocardium to the apex and back down to the lateral annulus.

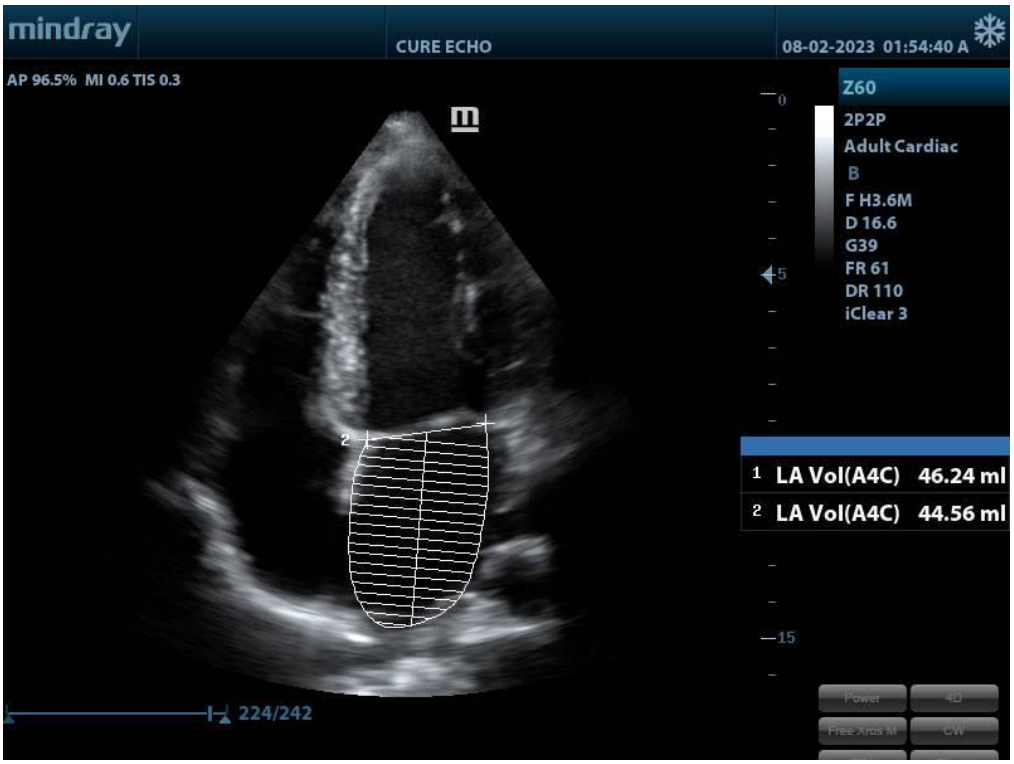
*Measure, Simpsons, ESV(a4C)



36. Simpson's Left atrial volume

Note: While the LV is at an end systolic state, use LA Simpson's to trace the LA volume. Start at the medial annulus and plot points down and around to the medial annulus.

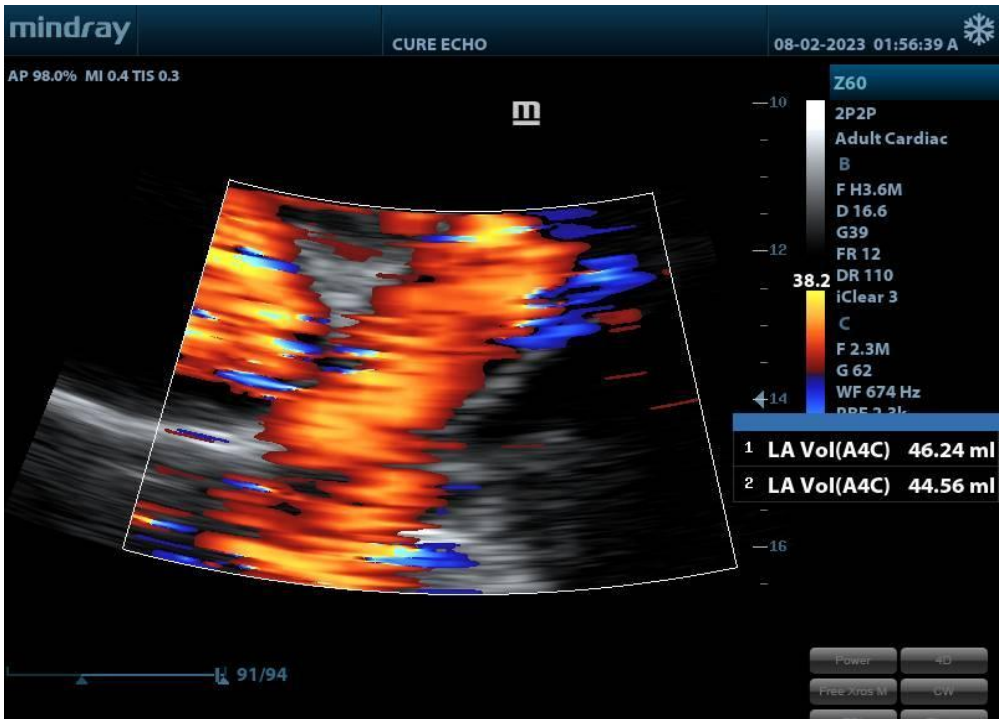
*measure, volume , LA Volume



37. Zoom and color pulmonary veins clip.

Note: Image should be zoomed over the pulmonary veins (bottom of LA). Color should demonstrate blood into the LA.

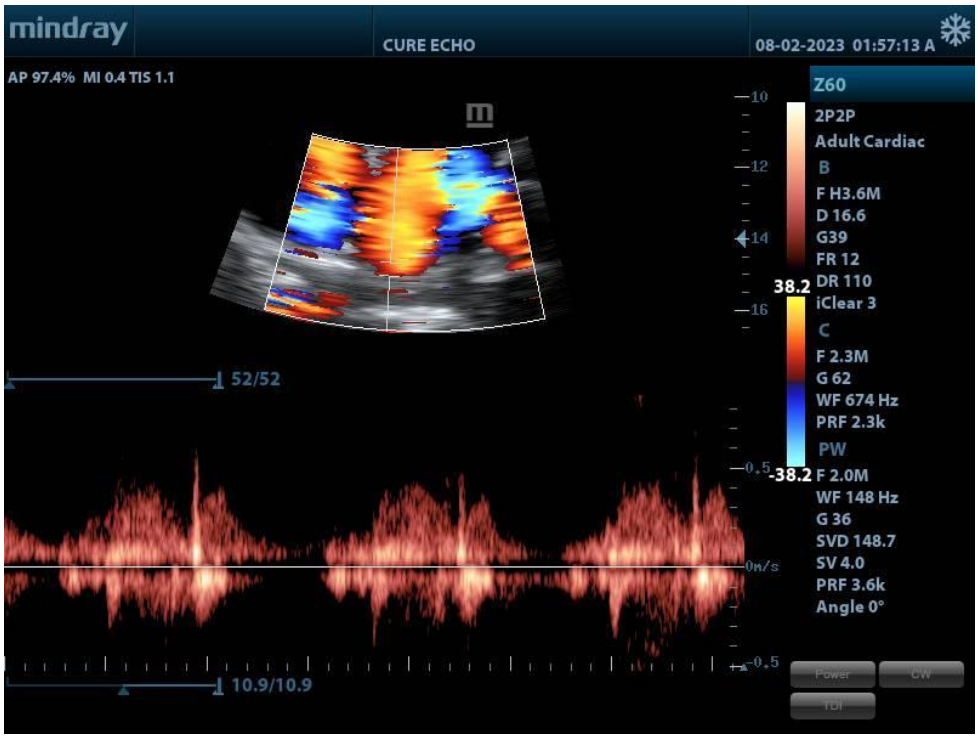
*To zoom press the center of the depth wheel and adjust box size then press again



38. PW Pulmonary veins (zoom)

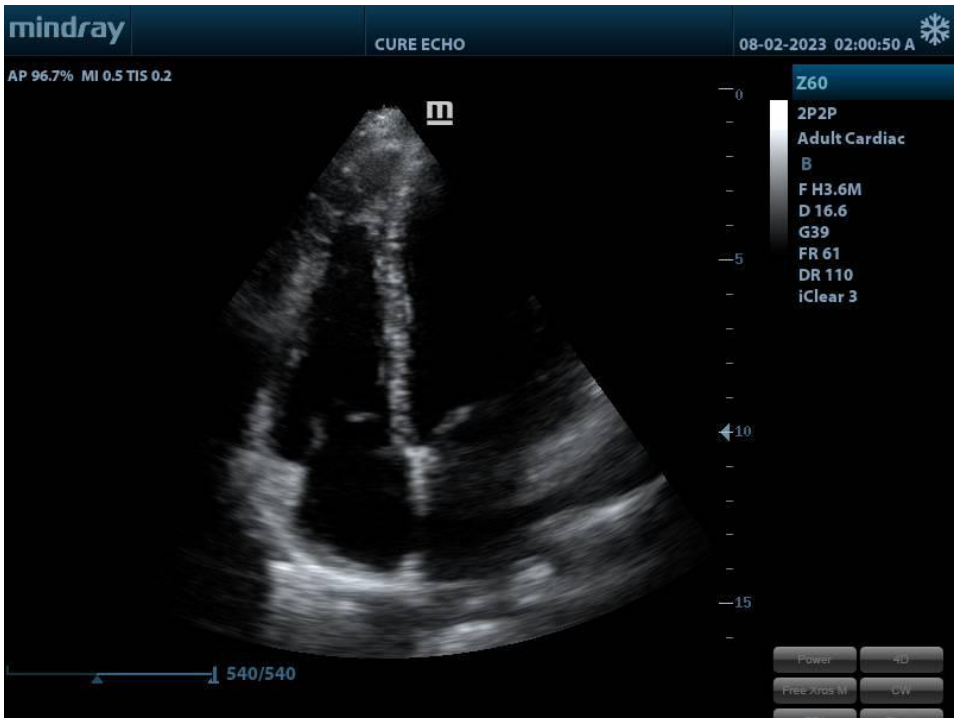
Note: Place sample volume at the junction of the pulmonary veins and the left atrium. Systolic and diastolic wave form should be seen above the baseline.

*PW button, place sample volume in pul. vein, pw to begin doppler



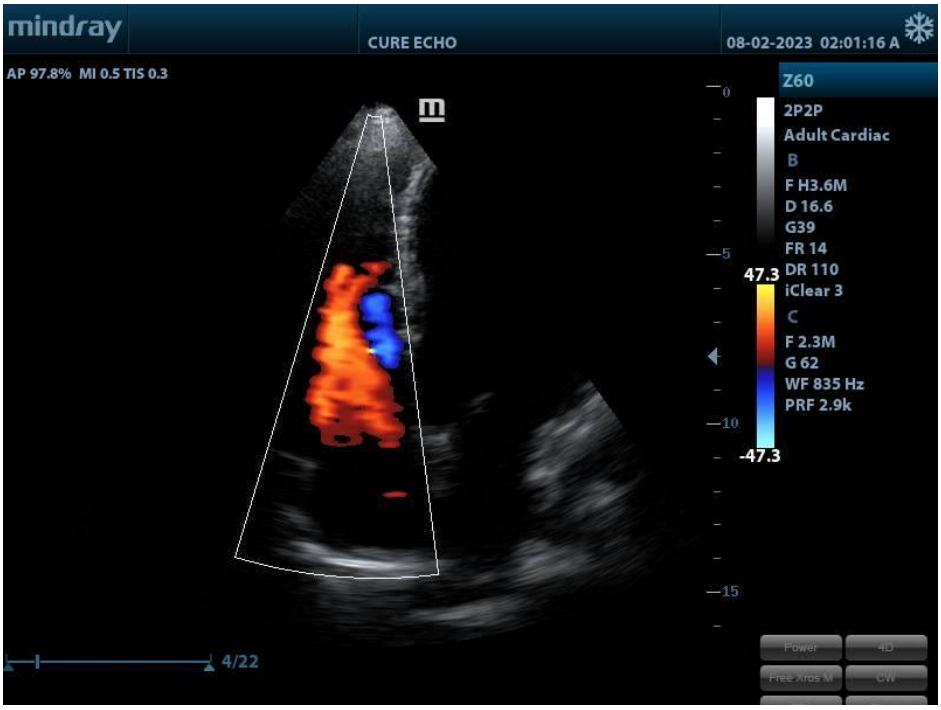
39. Right ventricular focus view clip

Note : This image should be focused on the right side of the heart. The IVS should remain parallel to the transducer. Optimize RV to assess RV function.



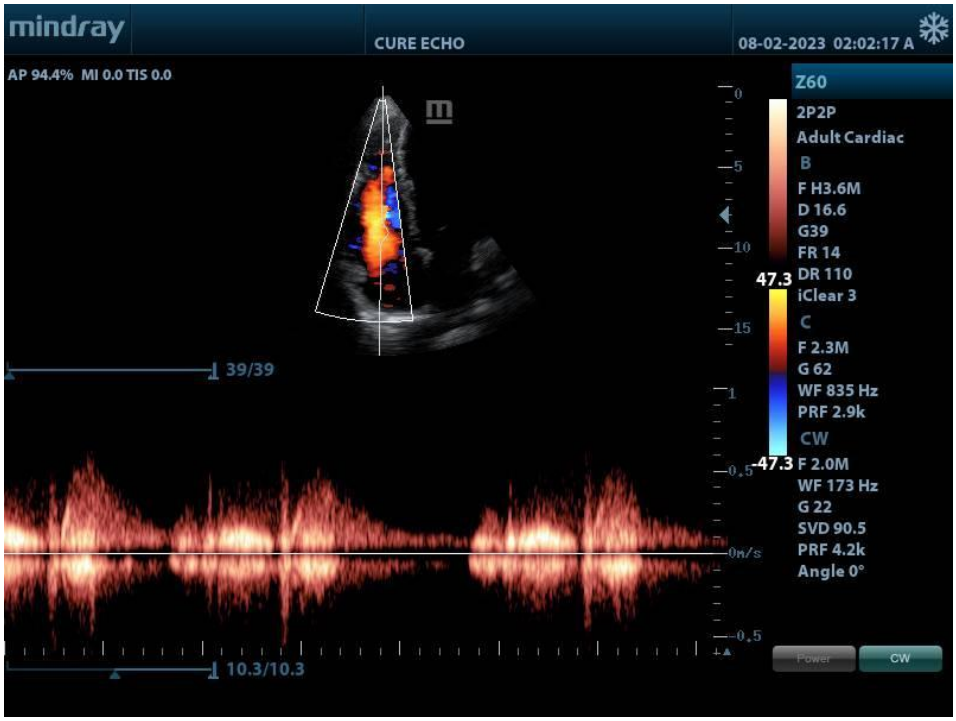
40. Color right ventricular inflow tract Apical 4 chamber

Note: Place color box over the right atrium and right ventricle. Normal RVIT flow will be positive (towards the transducer) and regurgitation will appear as flow away from the transducer.



41. Continuous wave right ventricular inflow tract

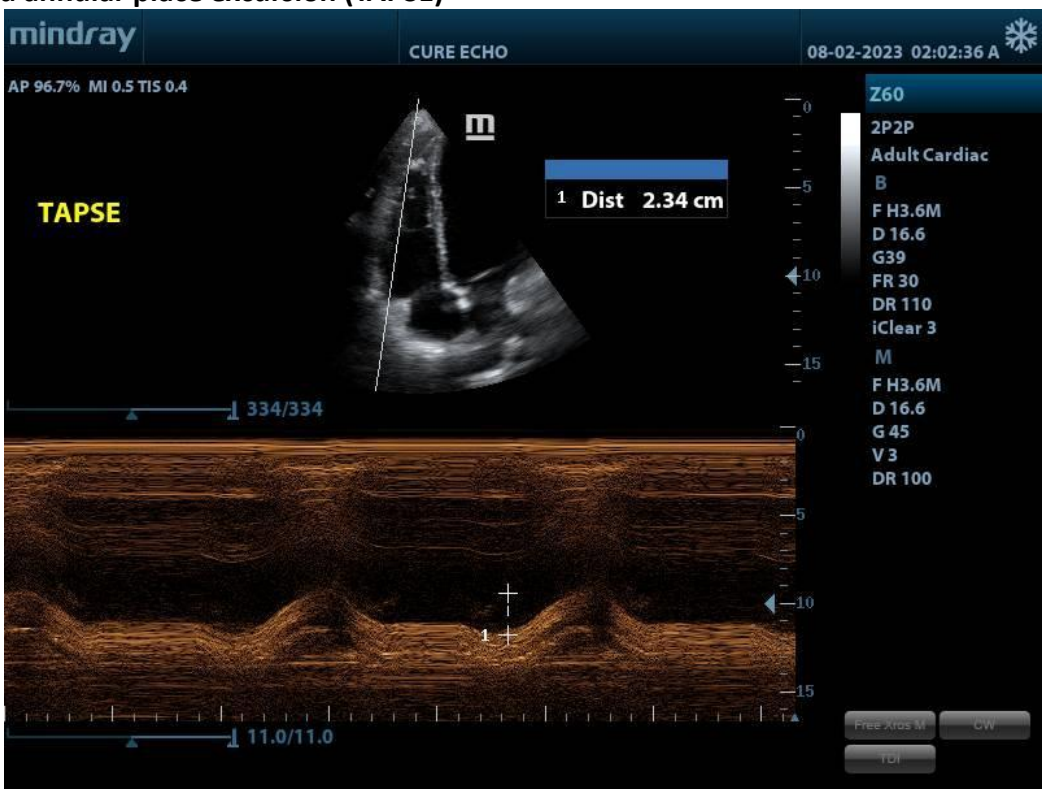
Note: CW doppler cursor placed parallel to RVIT flow. Normal RVIT flow will demonstrate a positive doppler shift. *Hit the print button to deploy CW cursor, align and hit print to begin doppler



42. Tricuspid annular place excursion (TAPSE)

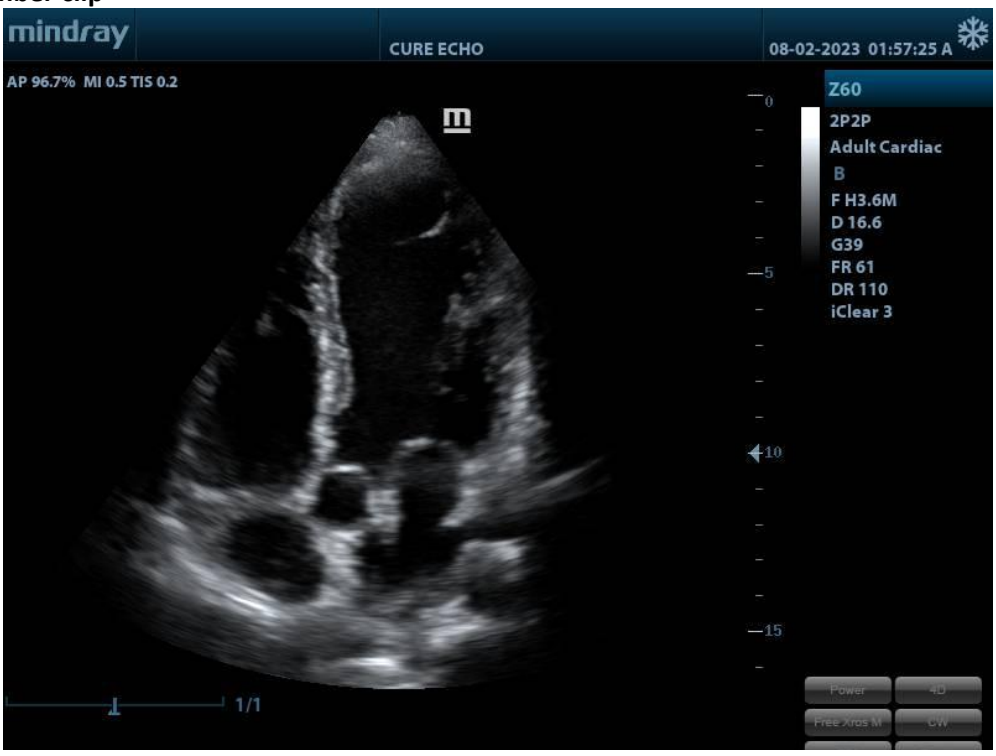
Note : M-mode cursor placed at the tricuspid valve annulus of the RV free wall. M-mode should demonstrate systolic and diastolic movement of the annulus. Measurement is taken from the lowest and highest points of annular motion.

*Caliper measurement



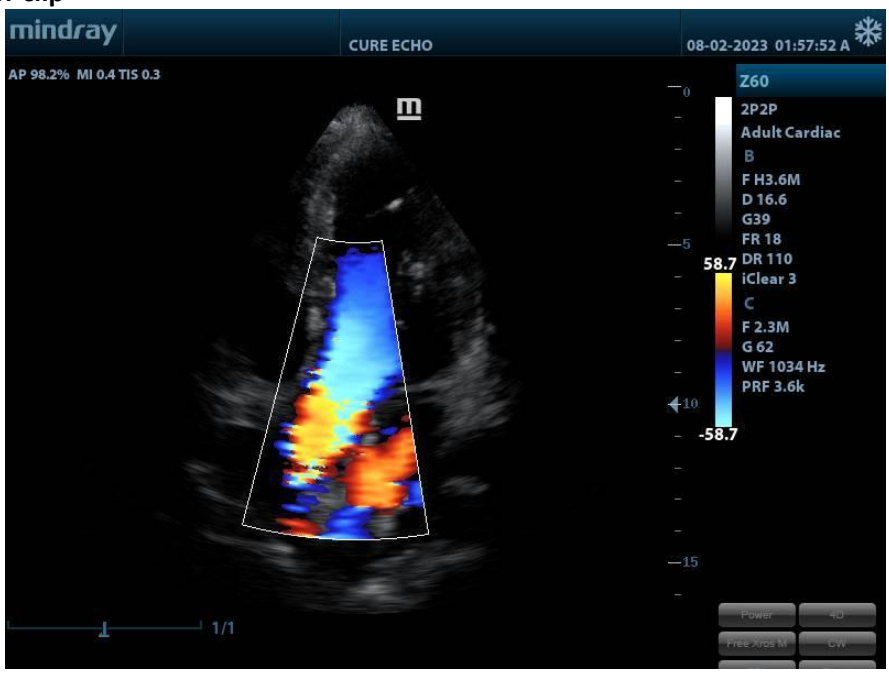
43. Apical 5 Chamber clip

Note : With notch remaining at 3oclock , angle the probe superiorly to open the LVOT and aortic valve.



44. Apical 5 Chamber color clip

Note : Place color box over the left ventricular inflow tract and aorta.

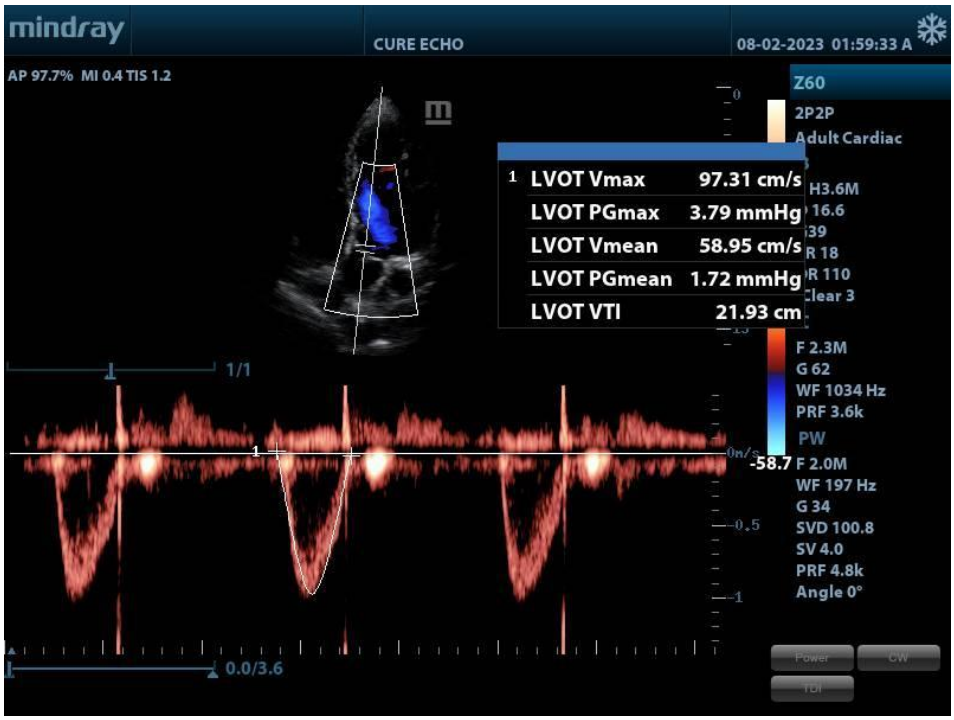


45. Pulsed wave left ventricular outflow tract

Note : Using PW , place the sample volume superiorly to the aortic valve in the LVOT.

*PW button, place sample volume in LVOT, PW to begin doppler

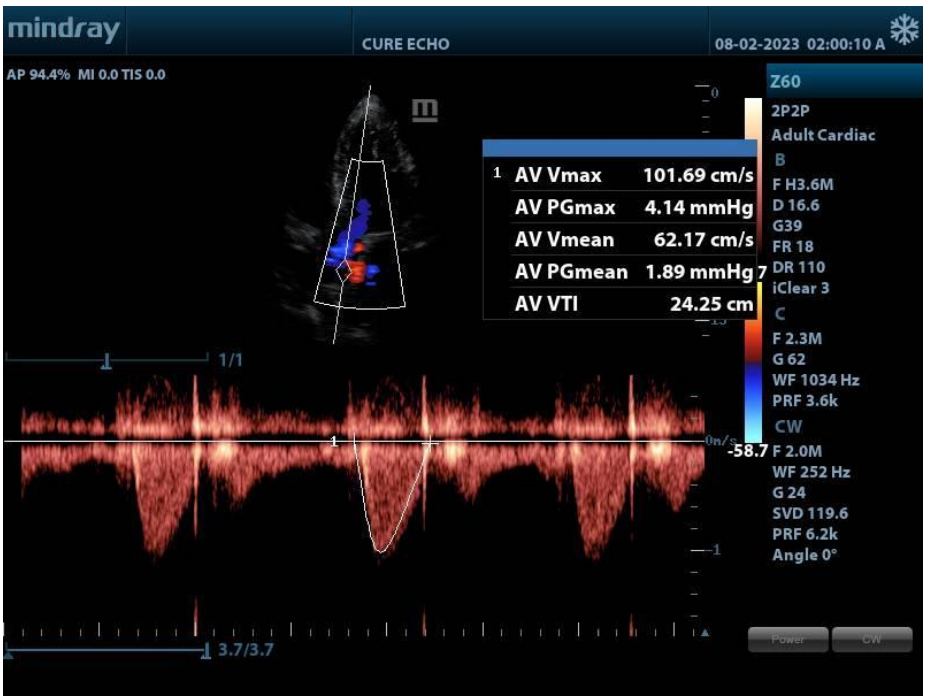
*Measure, aortic valve, LVOT VTI and trace waveform.



46. Continuous wave aortic valve

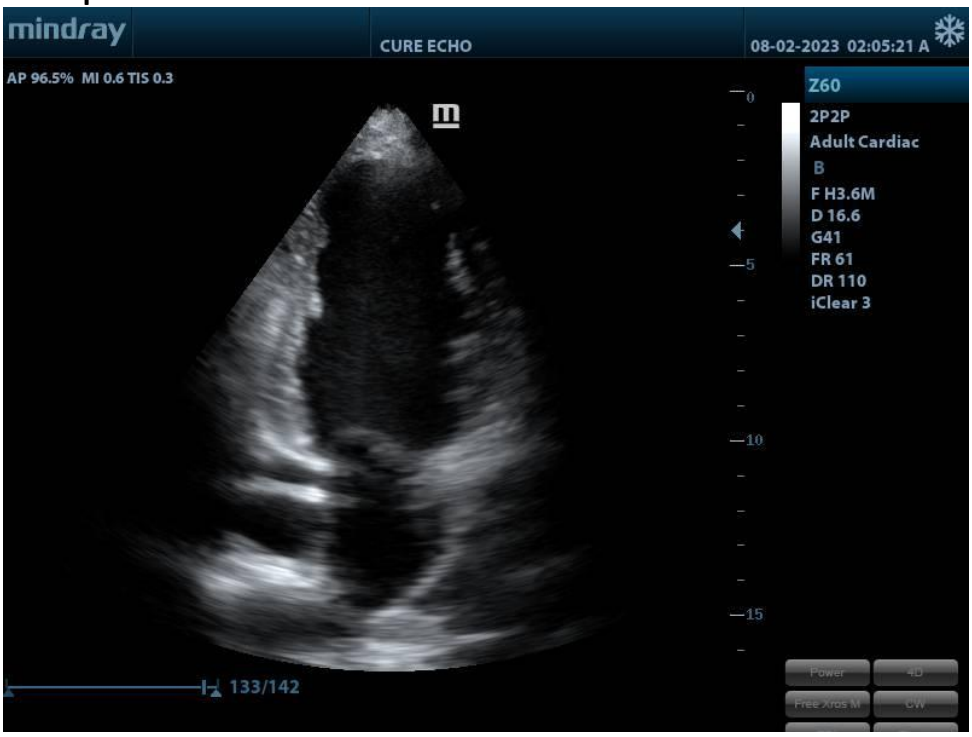
Note : Using CW place the CW focal point inferior to the aortic valve leaflets. It is important to make sure your cursor is parallel to aortic flow.
*Hit the print button to deploy CW cursor, align and hit print to begin doppler

*Measure, aortic valve, AOV VTI, trace waveform



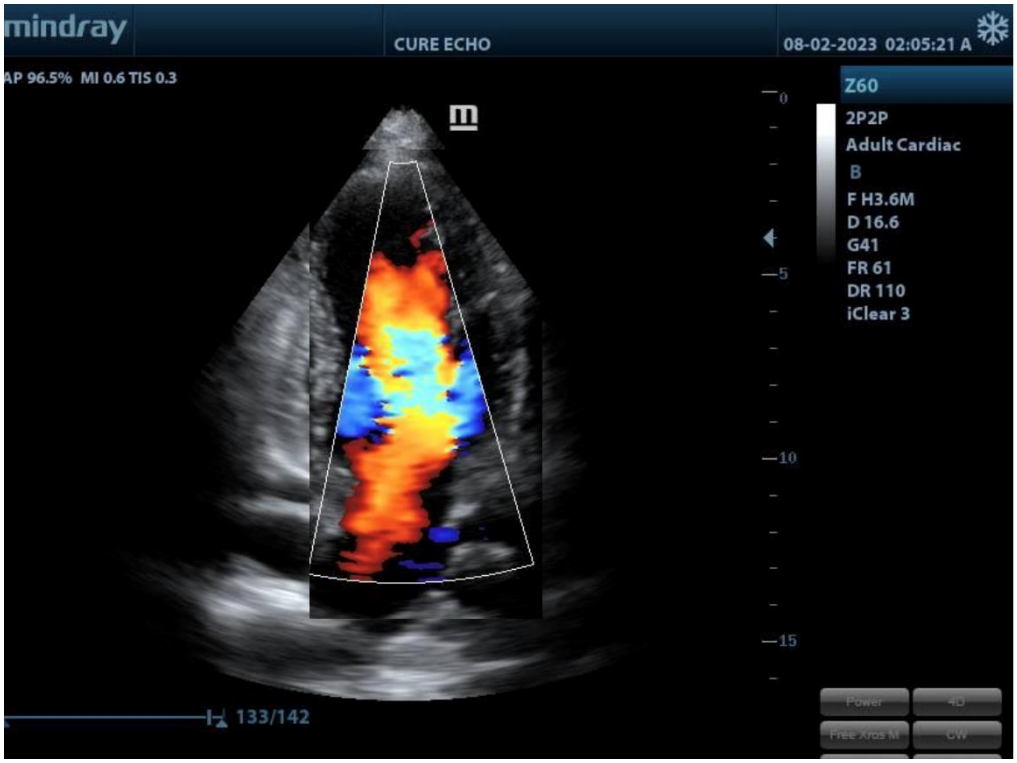
47. Apical 2 chamber clip

Note : While remaining In the same intercostal window , the probe notch should be rotated counterclockwise so that the notch moves from 3 o'clock to 12 o'clock . This two chamber view should only show the left ventricle and left atrium.



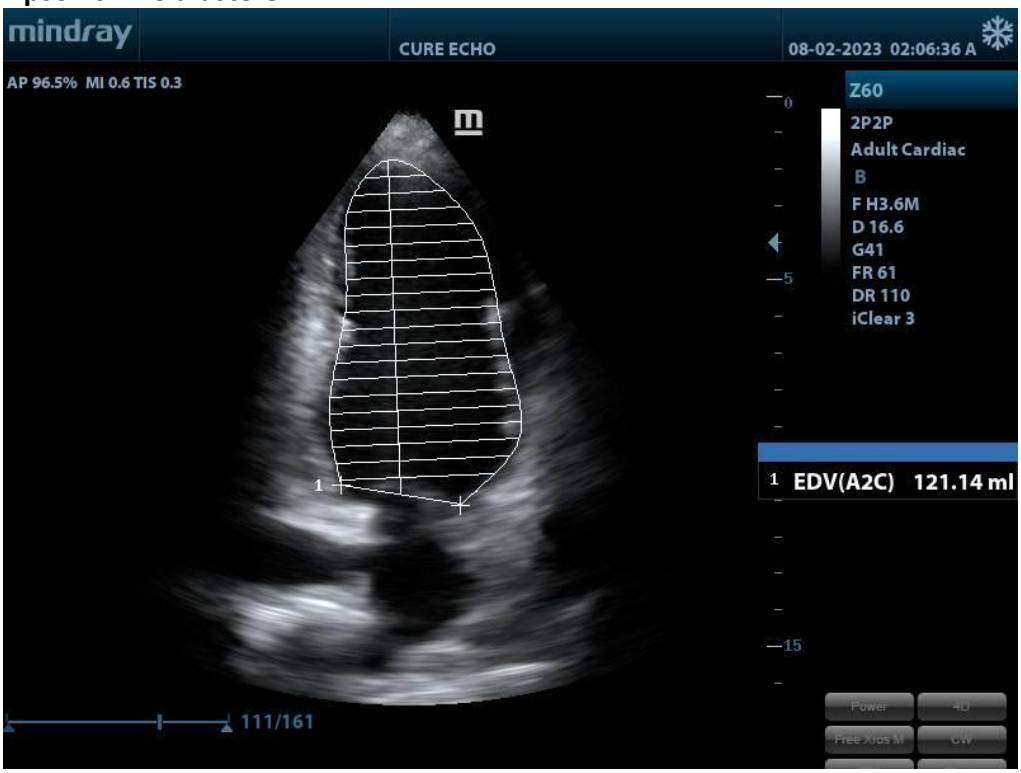
48. Apical 2 chamber color clip

Note: Color box should be placed over the LVIT.



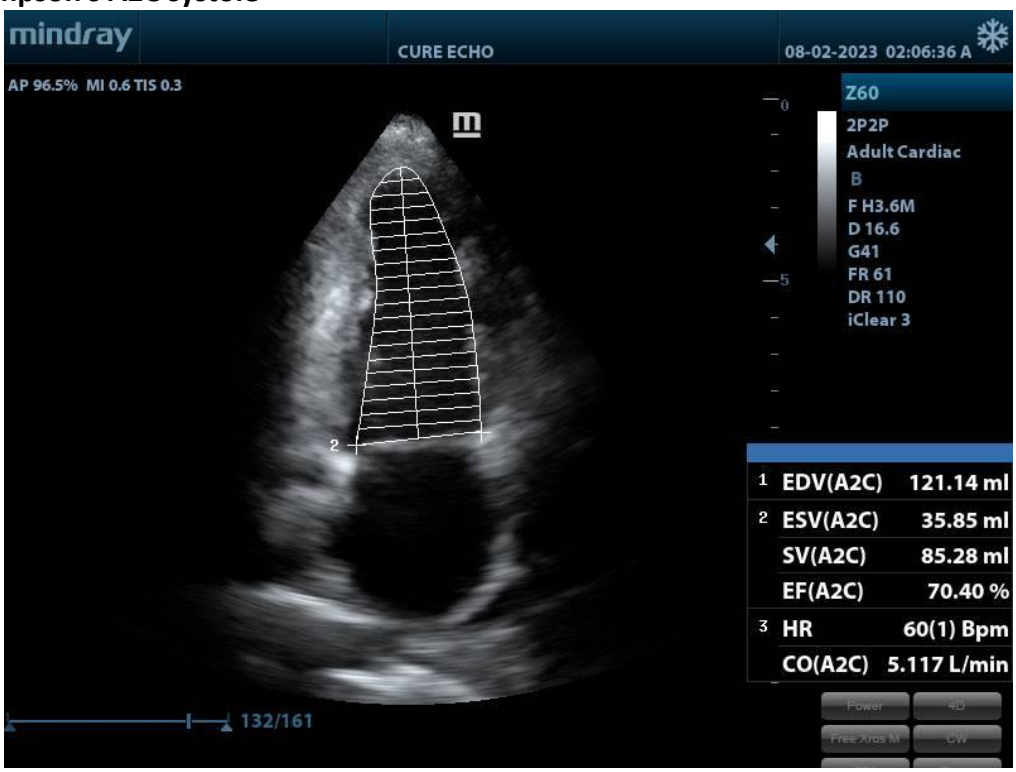
49. Biplane Simpson's A2C diastole

Note: Using the Left ventricle is in an end diastolic state, Simpsons measurements should be take. Starting at the annulus (inferior wall) and plotting points upwards towards the apex and back to the annulus (anterior wall).



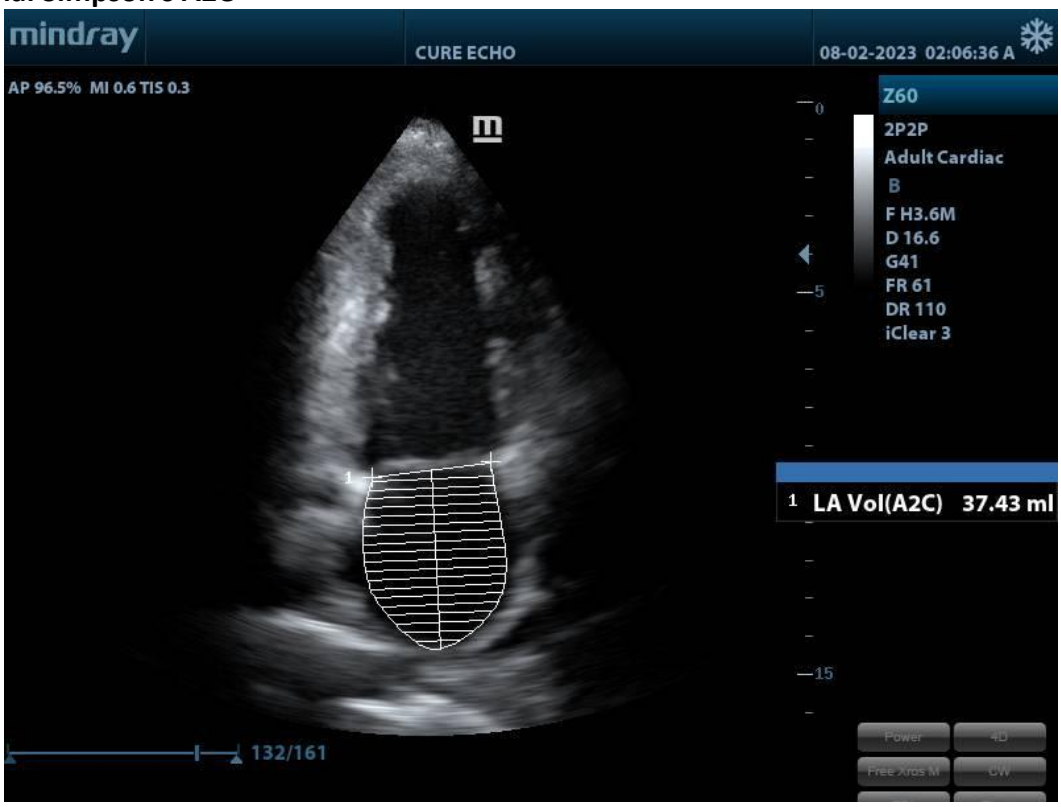
50. Biplane Simpson's A2C systole

Note: With the left ventricle in an end systolic state, Simpsons measurements should be take. Starting at the annulus (inferior wall) and moving up towards the apex and back down to the other annulus (anterior wall)



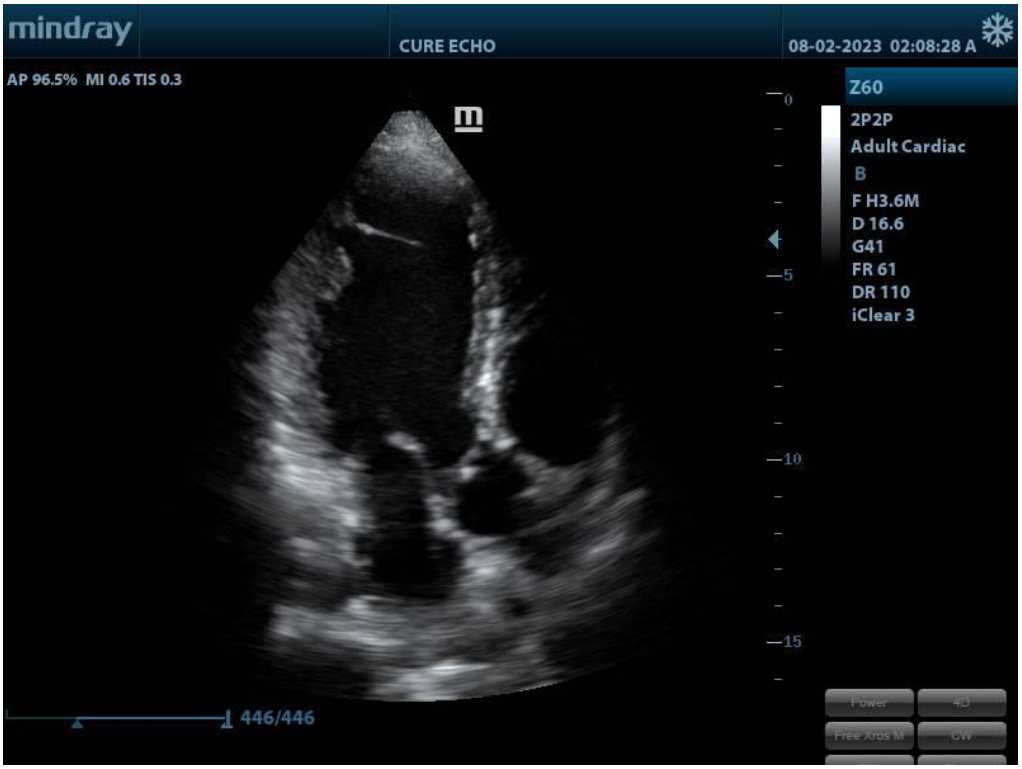
51. Left atrial Simpson's A2C

Note: With the left ventricle in an end systolic state the left atrial Simpsons measurement should be taken.



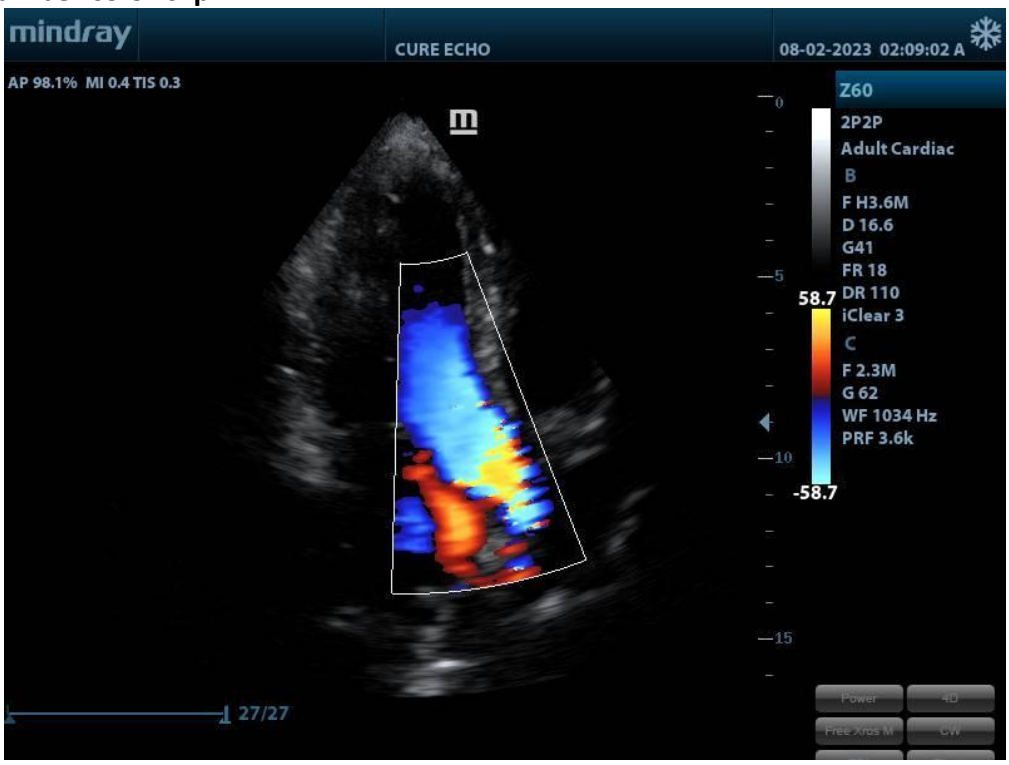
52. Apical 3 Chamber clip

Note : Rotate the probe counter clockwise to 11 o'clock. This will open the aorta.



53. Apical 3 chamber color clip

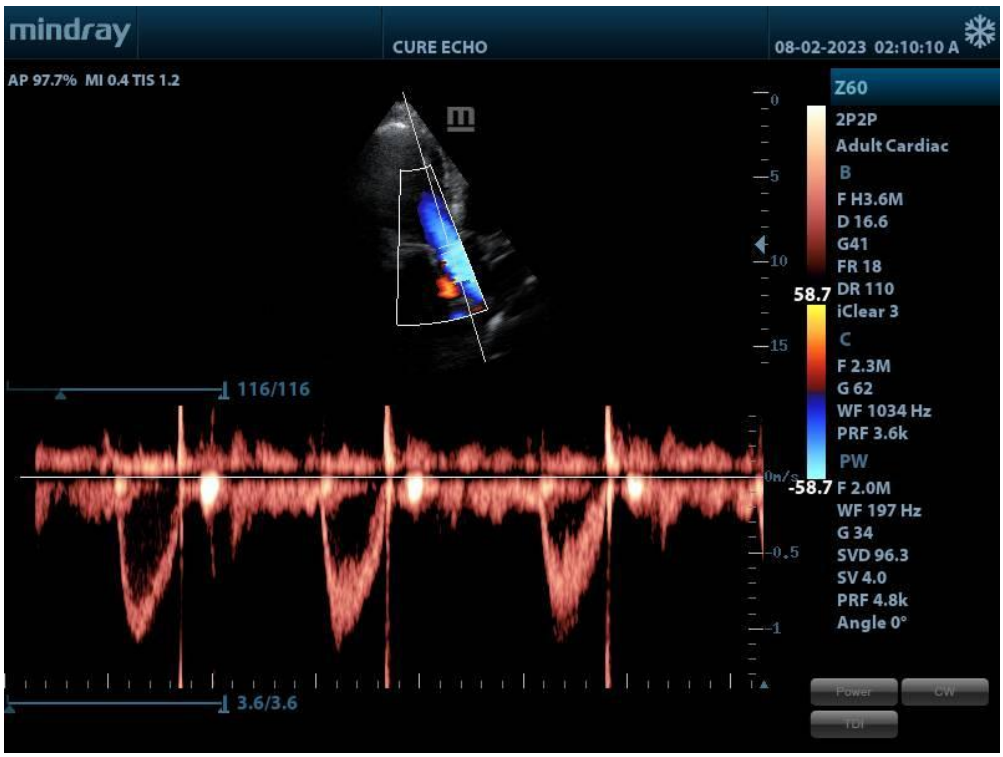
Note : Color box should be placed over the LVOT and Aortic valve.



54. Pulsed wave Left ventricular outflow tract

Note: Using PW , place the sample volume in the LVOT.

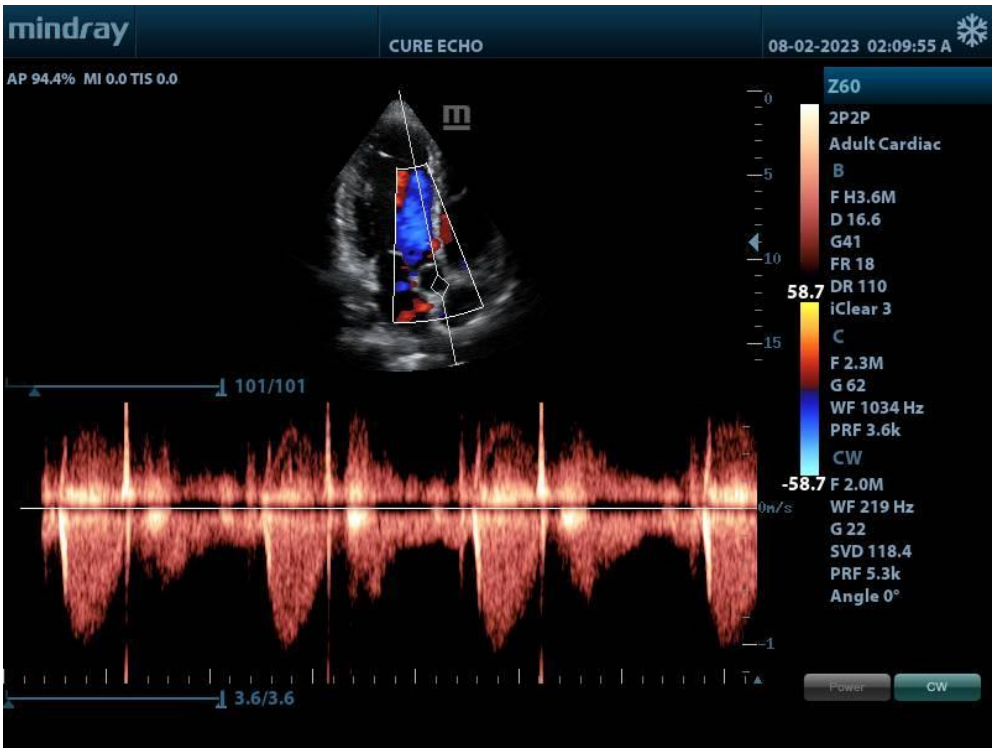
*PW, place sample volume in LVOT, PW to begin doppler



55. Continuous wave

Note : Using CW, place the cursor parallel to aortic flow with the focus in the aortic valve.

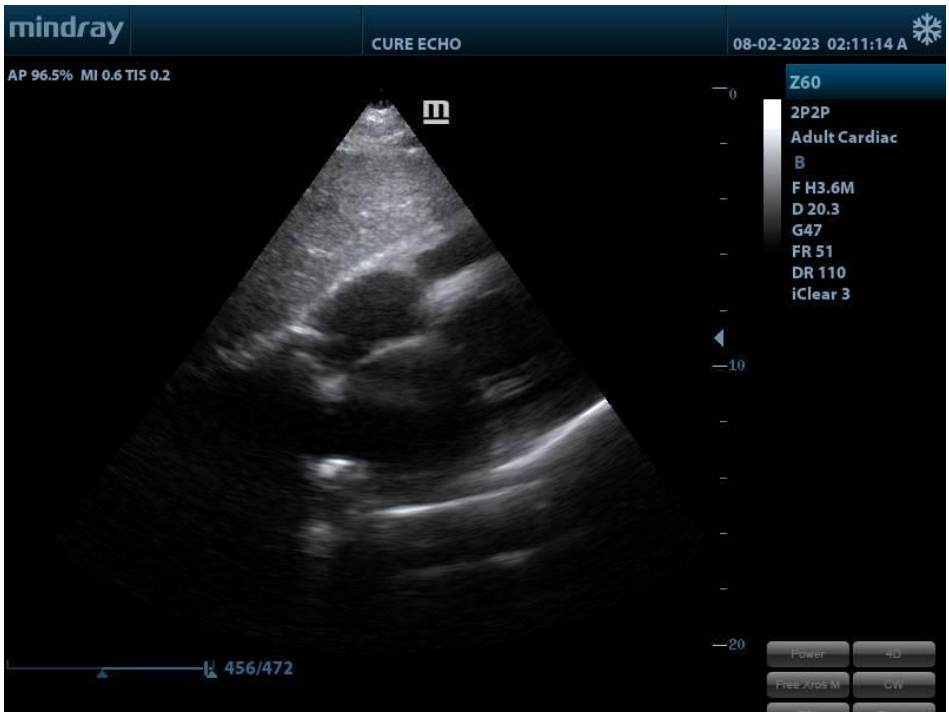
*Hit the print button to deploy CW cursor, align and hit print to begin doppler



Subcostal images: Patient should be transitioned to a supine position; Probe move to the xiphoid process with notch at 3 o'clock. Probe should be angled superiorly (back toward the heart).

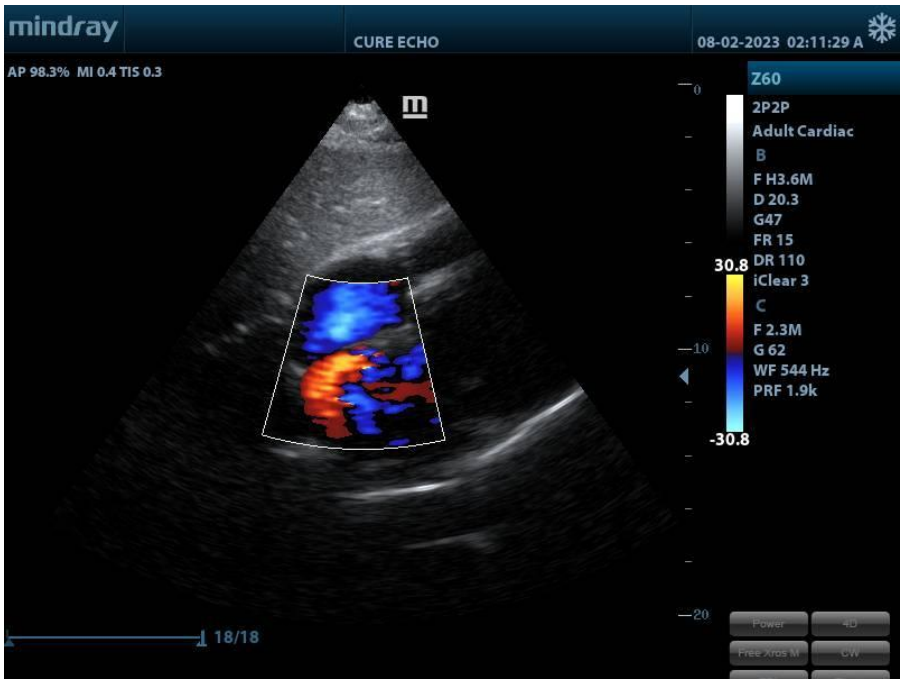
56. Subcostal 4 chamber

Note: All 4 chambers of the heart should be visible as well as the liver.



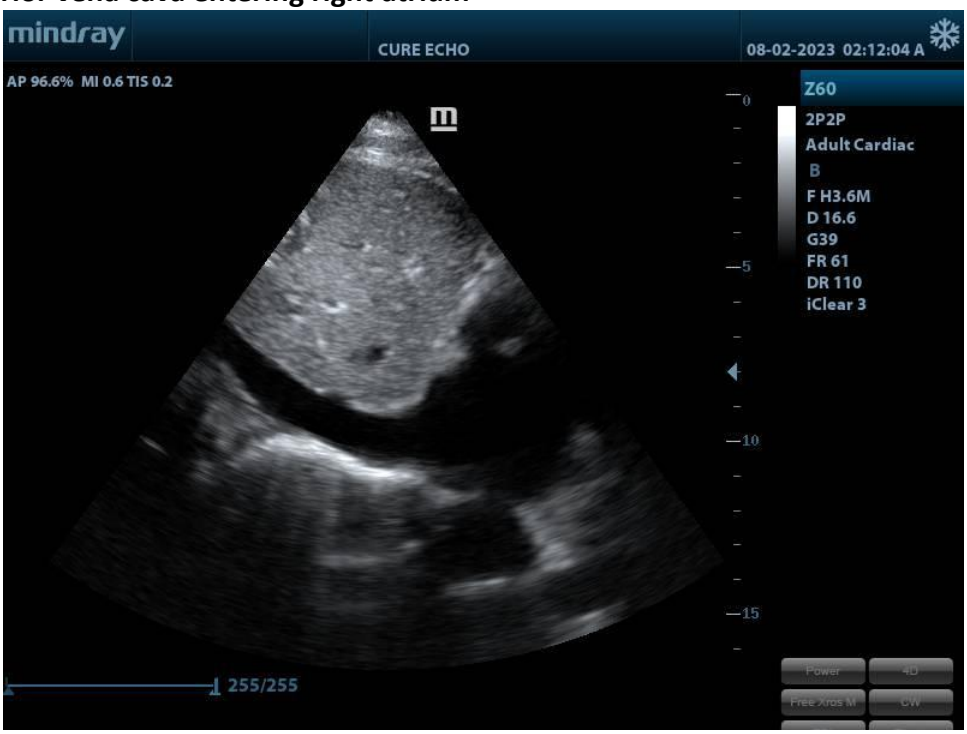
57. Color IAS 4chamber

Note: Color box should be focused on the interatrial septum. Make sure to lower color scale to allow for detection of low velocity shunts.



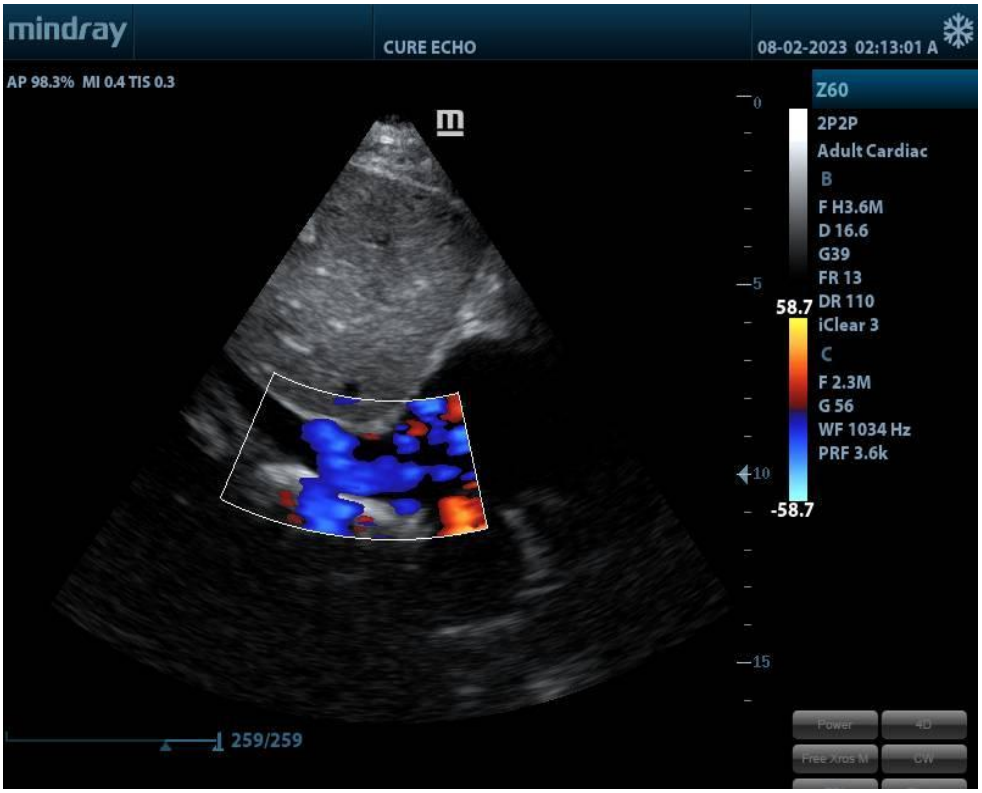
58. Liver with inferior vena cava entering right atrium

Note: Rotate the notch counterclockwise to the 12 o'clock position to open the inferior vena cava. Image should demonstrate the inferior vena cava entering the right atrium.



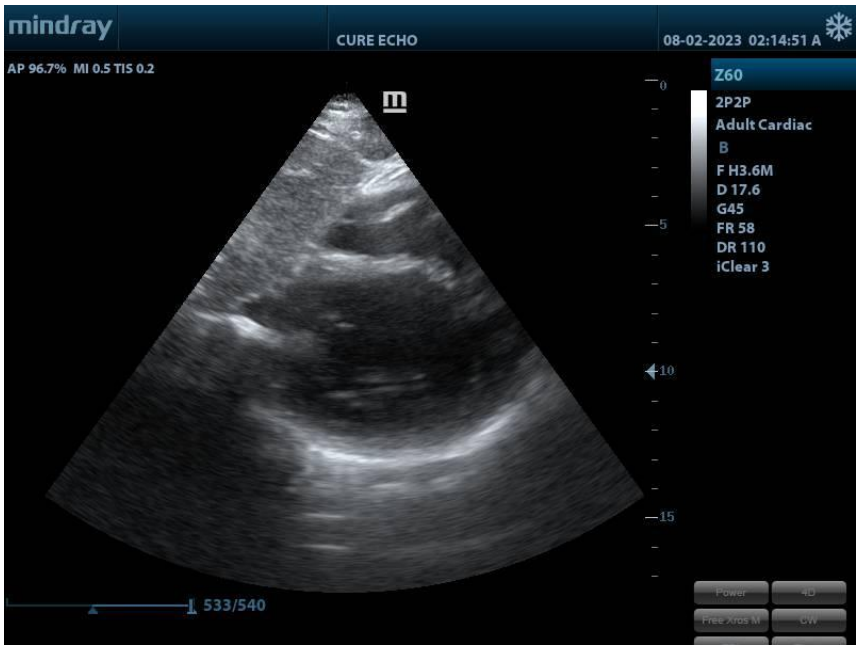
59. Color Inferior vena cava

Note: Color box should be placed over the proximal inferior vena cava and right atrium.



60. Subcostal short axis

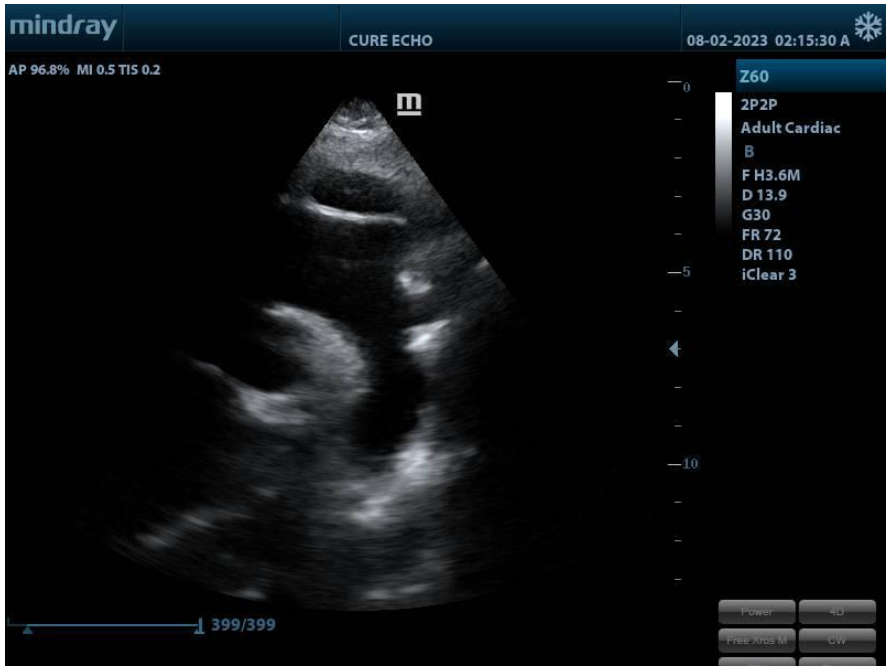
Note: To demonstrate a subcostal short axis view, the notch should be positioned at 11 o'clock. The LV will be seen in a cross-sectional view.



Aortic arch view: *Transducer should be placed in the suprasternal notch (dip between clavicle bones). The probe notch is in the 1oclock position with the probe angled towards the patient's left.*

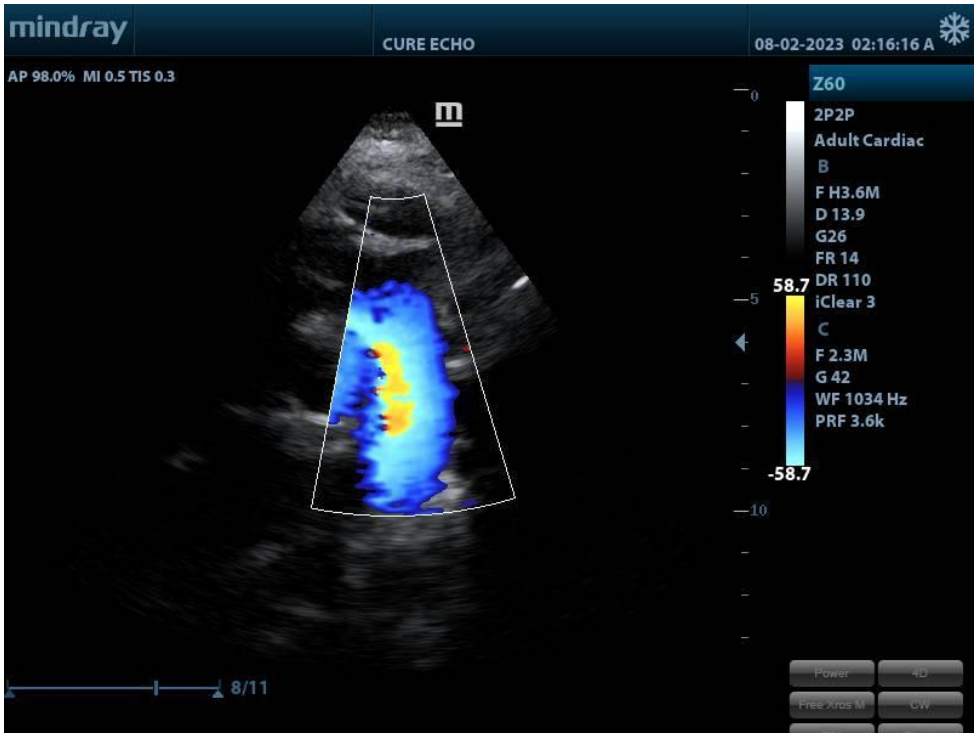
61. 2d aortic arch clip

Note : Aortic arch and branches should be clearly visible as well as a cross sectional view of the pulmonary artery.



62. Color aortic arch

Note : Color box should be placed over the aortic arch. Normal flow should be away from the transducer.



63. Continuous wave aortic arch

Note : Using CW , the doppler should be placed parallel to the descending thoracic aorta. Flow should be seen below the baseline (away from the transducer).

*Hit the print button to deploy CW cursor, align and hit print to begin doppler

